# Fisher forecasts from cached derivatives

How the derivatives are computed and stored, and why every FoM, error and shift is recomputed in seconds

November 2026

## 1 Summary

**The question.** The project's Fisher notebooks report figures of merit (FoMs), parameter errors and parameter shifts for four bias models, up to four linear-bias priors, seven scale cuts and four bin layouts, under dozens of assumed truths. One evaluation of the data vector runs CAMB, the code that computes the matter power spectrum, and CosmoLike, the code that projects it onto the sky for every thin bin; it takes seconds to tens of seconds. How, then, can a change of model, prior or scale cut cost only seconds?

**The answer.**

1. A Fisher forecast only needs the derivatives of the data vector at one point (Section 3).
2. The paper model is built from per-bin quantities – the biases, a counterterm and two stochastic amplitudes of every thin bin – tied to about twenty group-level parameters by calibrations, offsets and tilts (Section 4).
3. The base is one derivative column for each elementary knob: each cosmological parameter, each velocity dispersion, and each bias quantity in each thin bin. A model is a recipe that moves several knobs at once; written as a matrix $\mathsf{M}$, it turns the base $\mathsf{B}$ into the model's derivatives, $\mathsf{D} = \mathsf{BM}$, and a new model is a new $\mathsf{M}$ (Section 5).
4. Each layout's base is computed once – by the notebook, or by an MPI script on many cores – and stored on disk (Section 6).
5. A scale cut keeps some rows of the data vector. Each cut is reduced once to a small matrix $\mathsf{G}$ (Section 7), and every Fisher matrix, FoM and shift is then a product of small matrices (Section 8).
6. A derivative needs more numerical accuracy than a $\chi^2$, and a convergence study fixes the settings every published number uses (Section 9).
7. New evaluations are needed only for a term outside the base, a new fiducial point, or effects beyond first order (Section 10). Section 11 lists the files and when each must be recomputed.

**The setup.** The data are the DESI band powers $C_\ell^{gg}$: the angular power spectrum of galaxy clustering, averaged in 53 bands of multipole $\ell$. They are computed without the Limber approximation, which ties each multipole to a single wavenumber and fails for thin bins, for six tracer groups $g$ (BGS, LRG1, LRG2, LRG3, ELG1 and ELG2) on four layouts: bins of

| parameter | meaning | fiducial | prior |
|---|---|---|---|
| $10^9 A_s$ | primordial curvature amplitude | 2.1 | flat |
| $n_s$ | scalar spectral index | 0.96605 | flat |
| $H_0$ | Hubble constant [km/s/Mpc] | 67.32 | flat |
| $\Omega_b$ | baryon density today | 0.04 | flat |
| $\Omega_m$ | matter density today | 0.3 | flat |
| $w_0$ | dark-energy equation of state today | $-1$ | flat |
| $w_a$ | its slope, $w(a) = w_0 + w_a(1-a)$ | 0 | flat, as $w_0 + w_a$ |
| $\sum m_\nu$ | neutrino mass sum [eV] | 0.06 | fixed in the Fisher |
| $\tau$ | optical depth to reionization | 0.0697186 | fixed |

Table 1: The fiducial cosmology: the evaluate `override` of the paper yamls. The yamls' `ref` entries (chain starting points) differ from it and are not used.

equal comoving width $\Delta d_{\rm com} = 62$ Mpc, and bins of fixed redshift width $\Delta z = 0.02$, 0.05 and 0.2. Bin $i$ has mean redshift $\bar{z}_i$, and $D(z)$ is the linear growth factor, $D(0) = 1$. The model being forecast is the paper model: a one-loop bias model with per-group offsets and tilts (defined in the higher-order note, `notes/higher_order_bias`) under the pessimistic linear-bias prior of the correlated-prior note, `notes/correlated_bias_prior_v2`, which allows correlated wiggles of the bin biases around a straight line in redshift (Section 4). The machinery lives in one notebook, `EXAMPLE_EVALUATE_FISHER_MODEL.ipynb` ("the notebook" below), on the Fisher module `interface/cosmolike_lssty1xdesi_fisher.py`; the script `scripts/compute_fisher_base_mpi.py` computes the same derivative base on MPI ranks (Section 6).

## 2 Fiducial model and parameter values

This section freezes the point at which every derivative of this note was taken, so the note does not depend on files that may drift.

**The cosmology.** The fiducial point is the `override` block of the paper yamls `EXAMPLE_EVALUATE1–4.yaml`, the same in all four (Table 1). The Fisher matrices vary the first seven parameters; the neutrino mass stays at 0.06 eV and $\tau$ is fixed. At this point $\sigma_8 = 0.8278$.

**The layouts.** The four paper yamls $n = 1$ to 4 carry the layouts $\Delta d_{\rm com} = 62$ Mpc and $\Delta z =$ 0.02, 0.05, 0.2 (the datasets `lssty1xdesi_desi_dr3` and `..._dr1_dz0p02`, `dz0p05`, `dz0p20`): 76, 90, 36 and 10 thin bins in the six groups BGS, LRG1, LRG2, LRG3, ELG1 and ELG2, each spectrum in 53 band powers of width $\Delta\ell = 10$ from $\ell = 15$ to 544. Table 2 gives the sizes.

**The bias fiducial.** Bin $i$ of group $g$ has $b_{1,i} = B_g / D(\bar{z}_i)$, with $D$ read off the linear power spectrum ratio at $k = 0.005$/Mpc and the group amplitudes

| | BGS | LRG1 | LRG2 | LRG3 | ELG1 | ELG2 |
|---|---|---|---|---|---|---|
| $B_g$ | 1.34 | 1.7 | 1.7 | 1.7 | 0.84 | 0.84 |

$b_{2,i}$ sits on the Lazeyras calibration [1], $b_2^{\rm L}(b_1) = 0.412 - 2.143\, b_1 + 0.929\, b_1^2 + 0.008\, b_1^3$, and $b_{s^2,i}$ on co-evolution, $-\frac{4}{7}(b_{1,i} - 1)$; at the fiducial every offset, tilt, bias residual, counterterm ($\alpha_{0,i}$,

$\alpha_{2,i}$, both in $h^{-2}\,\mathrm{Mpc}^2$) and stochastic amplitude ($\mathrm{SN}_{0,i}$ in units of the bin's Poisson power $1/\bar{n}_i$; $\mathrm{SN}_{2,i}$ in units of $f_{\mathrm{sat}}\sigma^2_{\mathrm{sat}}/\bar{n}_i$, with $f_{\mathrm{sat}}\sigma^2_{\mathrm{sat}} = 6.084$ for BGS and LRG and 0.841 for ELG, in $(\mathrm{Mpc}/h)^2$) is zero. The linear-bias prior is the pessimistic one: residual width $\sigma_g = 0.03$ in every group, correlation length $L = 0.05$ in redshift, charging only the wiggles around each group's free line (Section 7).

**The velocity dispersions.** Seven $\sigma_v$ groups (BGS split at $\bar{z} = 0.255$), each with a Gaussian prior centered on its fiducial; the paper widths are six times the exploration widths that model B and model B + tilts use:

| | BGS0 | BGS1 | LRG1 | LRG2 | LRG3 | ELG1 | ELG2 |
|---|---|---|---|---|---|---|---|
| fiducial [km/s] | 371 | 358 | 357 | 348 | 343 | 304 | 289 |
| paper width [km/s] | 870 | 630 | 420 | 300 | 300 | 240 | 180 |
| exploration width [km/s] | 145 | 105 | 70 | 50 | 50 | 40 | 30 |

The yaml priors are truncated at $\sigma_v = 0$; at the paper width the BGS0 truncation is severe (the truncated prior's mean is about 850 km/s, not the fiducial 371). The Fisher matrices treat every $\sigma_v$ prior as an untruncated Gaussian of the quoted width, centered on the fiducial; these widths are what the derivative caches store.

**The model variants on this point.** The paper model is model B + tilts without $\alpha_2$, on the paper $\sigma_v$ widths. Model B and model B + tilts (the variants of the $k_{\mathrm{max}}$ scan) use the exploration $\sigma_v$ widths, free $\alpha_{2,g}$ per group, and the Gaussian widths 2 ($\Delta_{2,g}$), 1 ($\Delta_{s,g}$), 2 ($\mathrm{SN}_{0,g}$) and 5 ($\mathrm{SN}_{2,g}$); the counterterm widths are $0.5/k^2_{\mathrm{max}}$ at each cut, and every tilt is flat. All variants run under the baseline ($\sigma_g = 0.03$ for BGS, 0.01 otherwise; $L = 0.2$), pessimistic or free linear-bias prior.

**The scale cuts.** The non-Limber ultraviolet-fraction criterion of Saraivanov et al. [3] at $\alpha = 0.05$, bin $i$ cut at $k_{\mathrm{max}}/D(\bar{z}_i)$ with $k_{\mathrm{max}}$ quoted at $z = 0$, intersected with the shipped mask ($k = 0.1185/\mathrm{Mpc} = 0.176\,h/\mathrm{Mpc}$ in every bin). The cuts are computed on the paper fiducial above, at the reference settings of the derivative convergence study (every accuracy control above its MCMC value), for $k_{\mathrm{max}} = 0.05, 0.075, \ldots, 0.25\,h/\mathrm{Mpc}$, and cached in `chains/EXAMPLE_EVALUATE_FISHER_MODEL_SCALE_CUTS.pkl`; figures and FoMs use $k_{\mathrm{max}} \leq 0.2\,h/\mathrm{Mpc}$.

**The covariance.** Gaussian, computed at the one-loop fiducial above (with massless neutrinos, its one difference from the likelihood fiducial) for 14,000 square degrees; every Fisher matrix rescales its inverse by $f_A = 12{,}300/14{,}000 \approx 0.879$ to the survey area.

**The derivative tier.** Every number of this note is at the final tier, on every layout: CosmoLike `accuracyboost` 3, `nonlimber_accuracyboost` 64 (combined chi-grid factor $3 \times 64 = 192$, which the code rounds up to the power of two 256), `integration_accuracy` 1, `kmax_boltzmann` 20/Mpc (so the CAMB transfer reaches 60/Mpc), the low-rank engine's rank 5 and multipole cap 1000, CAMB `AccuracyBoost` 2 and `k_per_logint` 30, with the halofit tolerance patch. Five-point stencils everywhere, never central differences: relative step $h = 0.02$ for the sampled parameters, per-bin operator steps 0.01 ($b_{1,i}$, relative), 0.02 ($b_{2,i}$ and $b_{s^2,i}$, absolute) and $5\,h^{-2}\,\mathrm{Mpc}^2$ ($\alpha_{0,i}$); the linear $\alpha_{2,i}$, $\mathrm{SN}_{0,i}$ and $\mathrm{SN}_{2,i}$ columns are exact (Section 6.3). Section 9 is the evidence.

## 3 Why derivatives at one point are enough

Let $m(p)$ be the expected data vector at the parameters $p$, and $p_0$ the fiducial point, the parameter values around which the forecast is made. The vector $m$ has $n$ entries: the band powers of every bin's auto spectrum and of every measured cross spectrum between two bins, after 795 entries that the data format reserves for cosmic-shear blocks and that stay zero for DESI alone. $\mathsf{C}$ is the covariance of the data: how the measured band powers scatter around $m$ and how their errors correlate.

Near $p_0$ the data vector is close to a straight line in the parameters,

$$m(p) \;\simeq\; m(p_0) + \mathsf{D}\,(p - p_0), \qquad \mathsf{D}_{rj} = \left.\frac{\partial m_r}{\partial p_j}\right|_{p_0}, \tag{1}$$

with $r$ one row, that is one band power. With Gaussian data errors and Gaussian priors, the probability of the parameters given the data is then a Gaussian in $p$, whose inverse covariance is the Fisher matrix

$$F \;=\; \mathsf{D}^{\mathsf{T}}\mathsf{C}^{-1}\mathsf{D} \;+\; F_{\text{prior}}. \tag{2}$$

$F_{\text{prior}}$ is the priors' contribution: a Gaussian prior of width $w$ adds $1/w^2$ to its parameter's diagonal entry, and a flat prior adds nothing.

The inverse $F^{-1}$ is the forecast covariance of the parameters. $\sigma_j = [(F^{-1})_{jj}]^{1/2}$ is the marginalized error of parameter $j$, its error when every other parameter is free. $\mathrm{FoM}(w_0, w_a) = [\det \mathsf{C}_w]^{-1/2}$ is the inverse area of the $(w_0, w_a)$ error ellipse, with $\mathsf{C}_w$ the $(w_0, w_a)$ block of $F^{-1}$ and $w(a) = w_0 + w_a(1 - a)$ the dark-energy equation of state ($a$ the scale factor). $\mathrm{FoM}(\Omega_m, 10^9 A_s)$ is defined the same way, with $\Omega_m$ the matter density today in units of the critical density and $A_s$ the amplitude of the primordial curvature power spectrum.

The model enters $F$ only through $\mathsf{D}$, at one point. The covariance is Gaussian, computed at the one-loop fiducial (the fiducial point of the one-loop bias model, Section 6.2) for 14,000 square degrees, with its inverse rescaled by $f_A = 12{,}300/14{,}000$ to the survey area: a Gaussian covariance is inversely proportional to the area (more sky, more independent modes, smaller errors), so the smaller survey multiplies the covariance by $1/f_A$ and its inverse by $f_A \approx 0.879$ (`invcov_scale` in the code and the caches). It does not depend on the model being fitted. Its own numerical accuracy was audited when it was built: its non-Limber distance grid sits where a finer grid moves the adjacent-bin cross band powers by less than 0.4%, while the next coarser grid misses them by up to 15–30% (`covariance/README.md` records the measured convergence).

## 4 The model being forecast

**The quantities, bin by bin.** Everything the notebooks fit is built from a small set of quantities:

- seven cosmological parameters, with flat priors: $10^9 A_s$, the spectral index $n_s$, the Hubble constant $H_0$, the baryon density $\Omega_b$, $\Omega_m$, $w_0$ and $w_a$;
- seven velocity dispersions $\sigma_{v,1}, \ldots, \sigma_{v,7}$, with Gaussian priors, one per set of bins that shares a value. They smooth each bin's redshift distribution for the random velocities of galaxies (the Fingers-of-God effect). The sets follow the six tracer groups, except that

BGS takes two: on $\Delta d_{\rm com} = 62$ Mpc the dispersions cover bins 1–10 and 11–18 (BGS), 19–29 (LRG1), 30–38 (LRG2), 39–49 (LRG3), 50–61 (ELG1) and 62–76 (ELG2);

- separately in every thin bin $i = 1, \ldots, N$: the linear bias $b_{1,i}$; the quadratic and tidal biases $b_{2,i}$ and $b_{s^2,i}$, the galaxies' responses to the squared density and to the tidal field; the counterterm $\alpha_{0,i}$, a term proportional to $k^2$ that absorbs small-scale physics; and the stochastic amplitudes $\mathrm{SN}_{0,i}$ and $\mathrm{SN}_{2,i}$, noise-like terms from the discreteness of galaxies.

**The per-bin power spectrum.** The real-space galaxy power spectrum of bin $i$ is

$$P_{gg,i}(k) \;=\; b_{1,i}^2\, P_{\rm mm}(k) \;+\; P_{\rm loop}(k;\, b_{1,i}, b_{2,i}, b_{s^2,i}) \;+\; \alpha_{0,i}\, k^2 P_{\rm mm}(k), \tag{3}$$

with $P_{\rm mm}$ the nonlinear matter power spectrum and $P_{\rm loop}$ the one-loop terms, the next order of perturbation theory beyond linear, which carry $b_{2,i}$ and $b_{s^2,i}$. $\alpha_{0,i}\, k^2 P_{\rm mm}$ is a counterterm, in $h^{-2}\,\mathrm{Mpc}^2$: a term that absorbs small-scale physics the model does not compute. In redshift space the model adds the stochastic terms $\mathrm{SN}_{0,i}$ (a white noise beyond the Poisson one) and $\mathrm{SN}_{2,i}\, k^2\mu^2$ (its line-of-sight counterpart, with $\mu$ the cosine of the angle to the line of sight). Section 6.3 writes out how every term reaches the projected band powers; the full physics is in the higher-order note, `notes/higher_order_bias`, and the correlated-prior note, `notes/correlated_bias_prior_v2`.

**How the fit parameterizes the bins.** No fit gives every bin its own free quantities; the parameter files tie the bins to their groups:

- the linear bias follows one amplitude $B_g$ per group, $b_{1,i} = B_g / D(\bar{z}_i)$: the group's clustering amplitude $b_1 D$ is constant across its bins;
- $b_{2,i}$ follows $b_{1,i}$ through a calibration, the relation $b_2^{\rm L}(b_{1,i})$ fitted to simulations by Lazeyras et al. [1], plus one offset $\Delta_{2,g}$ per group; $b_{s^2,i}$ follows the co-evolution relation, the prediction for galaxies that form with no tidal bias and then move with the matter, $b_{s^2,i} = -\frac{4}{7}(b_{1,i} - 1) + \Delta_{s,g}$, with its own group offset;
- $\alpha_{0,i}$, $\mathrm{SN}_{0,i}$ and $\mathrm{SN}_{2,i}$ take one value per group: $\alpha_{0,g}$, $\mathrm{SN}_{0,g}$ and $\mathrm{SN}_{2,g}$;
- the tilts let a group value drift linearly in redshift: $\Delta_{2,g}$ becomes $\Delta_{2,g} + t_{2,g}(\bar{z}_i - \bar{z}_g)$, with $t_{2,g}$ the tilt and $\bar{z}_g$ the group's mean bin redshift, and likewise for $\Delta_{s,g}$ and $\alpha_{0,g}$; the stochastic amplitudes never tilt;
- the linear bias is fitted under the pessimistic prior of the correlated-prior note: one amplitude per bin, with a free amplitude and tilt per group and correlated residuals around them. (That note defines three more linear-bias priors, from growth-tied to free; the machinery below takes any of them.)

**The paper model.** The model this note forecasts is the paper model: the one-loop bias model above with, per group, the offsets $\Delta_{2,g}$ and $\Delta_{s,g}$, the counterterm amplitude $\alpha_{0,g}$, the stochastic amplitudes $\mathrm{SN}_{0,g}$ and $\mathrm{SN}_{2,g}$ and the three tilts, under the pessimistic linear-bias prior. Every group parameter carries the Gaussian prior width of Section 2, with all seven $\sigma_v$ widths six times the exploration fits'. The widened dispersions stand in for the redshift-space

counterterm $\alpha_2$ of the DESI DR1 convention [2], a $k^2\mu^2$ term the model sets to zero in every bin: $\alpha_2$ is about 99% degenerate with $\sigma_v$, and the widened $\sigma_v$ absorbs a DESI-size $\alpha_2$ at about the same FoM for $k_{\max} \geq 0.1\,h$/Mpc (the Fingers-of-God note, `notes/fog_parameter_degeneracy_study`). The same machinery builds the recipe matrix of any other model assembled from the same quantities, which the exploration notebooks fit and the correlated-prior and higher-order notes describe.

**Why the forecast needs per-bin quantities.** A chain samples only the group-level parameters above, about twenty in all. The forecast, though, must price choices that differ in exactly how the bins are tied together: one amplitude per group, per bin, or correlated in between; offsets with or without tilts. All of them are recipes on the same per-bin quantities, so the object worth precomputing is the response of the data vector to each quantity in each single bin. That is the base of the next section.

# 5 The base, and every model as a matrix on it

## 5.1 What the base is

The base is the list of elementary knobs the code turns by hand: each cosmological parameter, each velocity dispersion, and, separately in every thin bin, each of the six bias quantities of Section 4 plus the redshift-space counterterm $\alpha_{2,i}$ – seven per-bin families in all. The paper model fixes $\alpha_{2,i} = 0$, but model B frees it and the shift tests of Section 8 assume truths with it, so the base carries its columns too. For each knob, the code computes how every band power changes when that knob alone moves: one derivative column, a vector of $n$ numbers. These are the base columns, and they are what the notebook precomputes and stores (Section 8.1).

The base matrix $\mathsf{B}$ puts the base columns side by side. A model parameter, such as a group offset, a tilt or a per-bin amplitude, is a recipe that moves several knobs at once. The chain rule turns that recipe into a column of the model's derivative matrix: how $m$ changes with $p_j$ is the sum, over the knobs, of how $m$ changes with the knob times how the knob changes with $p_j$,

$$\frac{\partial m}{\partial p_j} = \sum_a \frac{\partial m}{\partial \kappa_a}\frac{\partial \kappa_a}{\partial p_j}, \qquad \text{that is} \qquad \mathsf{D} = \mathsf{B}\,\mathsf{M}, \qquad \mathsf{M}_{aj} = \frac{\partial \kappa_a}{\partial p_j}, \tag{4}$$

with $\kappa_a$ the value of knob $a$ and $p_j$ the model's parameter $j$. $\mathsf{M}$ is the recipes written as numbers: $\mathsf{M}_{aj}$ is how much knob $a$ moves per unit of parameter $j$. Section 5.2 works one out.

**There is no single $b_1$.** Every bias knob belongs to one bin. "The $b_1$ columns" are therefore $N$ columns, $\partial m/\partial b_{1,1}, \ldots, \partial m/\partial b_{1,N}$, and the same holds for each of the seven bias families $\{b_{1,i}\}$, $\{b_{2,i}\}$, $\{b_{s^2,i}\}$, $\{\alpha_{0,i}\}$, $\{\mathrm{SN}_{0,i}\}$, $\{\mathrm{SN}_{2,i}\}$ and $\{\alpha_{2,i}\}$, $i = 1, \ldots, N$. In order, $\mathsf{B}$ has

$$n_{\text{base}} = \underbrace{7}_{\text{cosmology}} + \underbrace{7}_{\sigma_v} + \underbrace{7N}_{7\ \text{families}\times N\ \text{bins}} + \underbrace{1}_{\text{check}} = 7N + 15 \tag{5}$$

columns; the check column is explained in Section 10. On $\Delta d_{\text{com}} = 62$ Mpc there are $N = 76$ bins, so $\mathsf{B}$ has $7 + 7 + 532 + 1 = 547$ columns of $n = 10{,}494$ numbers each. Table 2 gives every layout.

| layout | bins $N$ | cross spectra | entries $n$ | $n_{\text{base}}$ | kept, shipped mask |
|---|---|---|---|---|---|
| $\Delta d_{\text{com}} = 62$ Mpc | 76 | 107 | 10,494 | 547 | 5,674 |
| $\Delta z = 0.02$ | 90 | 133 | 12,614 | 645 | 7,621 |
| $\Delta z = 0.05$ | 36 | 52 | 5,459 | 267 | 2,982 |
| $\Delta z = 0.2$ | 10 | 13 | 2,014 | 85 | 744 |

Table 2: The sizes of one layout's Fisher problem. Cross spectra: the pairs of bins whose cross spectrum is in the data vector. $n$: entries of the data vector. $n_{\text{base}} = 7N + 15$: columns of the base. Last column: the band powers the shipped mask keeps (the mask of the data files, before any scale cut), the largest set of rows any cut uses.

### 5.2 A toy example

**The setup.** Take one group $g$ of $N = 3$ bins, with mean redshifts $\bar{z}_1$, $\bar{z}_2$, $\bar{z}_3$, their mean $\bar{z}_g$, and $D_i \equiv D(\bar{z}_i)$. Keep one band of each auto spectrum and of the two cross spectra of neighboring bins, so the data vector has $n = 5$ entries,

$$m = \left(C^{11}, C^{22}, C^{33}, C^{12}, C^{23}\right)^{\mathsf{T}}. \tag{6}$$

Keep one cosmological parameter $\theta$ and two bias families. The knobs are $\theta$, $b_{1,1}$, $b_{1,2}$, $b_{1,3}$, $b_{2,1}$, $b_{2,2}$ and $b_{2,3}$, so $n_{\text{base}} = 7$.

**The base matrix.** Name the base columns $y = \partial m/\partial\theta$, $u_i = \partial m/\partial b_{1,i}$ and $v_i = \partial m/\partial b_{2,i}$. Then

$$\mathsf{B} = \begin{array}{c} \\ C^{11} \\ C^{22} \\ C^{33} \\ C^{12} \\ C^{23} \end{array} \begin{array}{c} \begin{array}{ccccccc} y & u_1 & u_2 & u_3 & v_1 & v_2 & v_3 \end{array} \\ \begin{pmatrix} * & * & 0 & 0 & * & 0 & 0 \\ * & 0 & * & 0 & 0 & * & 0 \\ * & 0 & 0 & * & 0 & 0 & * \\ * & * & * & 0 & * & * & 0 \\ * & 0 & * & * & 0 & * & * \end{pmatrix} \end{array}, \tag{7}$$

where $*$ marks a nonzero entry. The column $u_1$ vanishes on $C^{22}$, $C^{33}$ and $C^{23}$, because the bias of bin 1 does not enter those spectra. A cross spectrum, such as $C^{12}$, depends on the biases of both its bins.

**The model.** The model has the parameters $p = (\theta, \Delta_{2,g}, t_{2,g}, \beta_1, \beta_2, \beta_3)$, so $n_{\text{par}} = 6$, and sets the knobs by two recipes:

$$b_{1,i} = \frac{\beta_i}{D_i}, \qquad b_{2,i} = b_2^{\text{L}}(b_{1,i}) + \Delta_{2,g} + t_{2,g}\,\zeta_i, \qquad \zeta_i \equiv \bar{z}_i - \bar{z}_g. \tag{8}$$

Here $\beta_i$ is the amplitude of bin $i$, and $b_2^{\text{L}}$ the calibration of Section 4, which predicts $b_{2,i}$ from $b_{1,i}$. $\Delta_{2,g}$ is the group's offset from that calibration and $t_{2,g}$ its tilt in redshift. Let $s_i \equiv b_2^{\text{L}\prime}(b_{1,i})$ be the slope of the calibration.

**The recipes as numbers.** $\mathsf{M}_{aj}$ is how much knob $a$ moves per unit of parameter $j$:

1. $\theta$ moves only $\theta$. ($y$ is computed by moving $\theta$ with the amplitudes $\beta_i$ held fixed, which is how the model holds them, so $\theta$ needs no chain rule.)

2. $\Delta_{2,g}$ moves $b_{2,1}$, $b_{2,2}$ and $b_{2,3}$ by 1 each.

3. $t_{2,g}$ moves $b_{2,i}$ by $\zeta_i$.

4. $\beta_i$ moves $b_{1,i}$ by $1/D_i$ and, through the calibration, $b_{2,i}$ by $s_i/D_i$; the other bins do not move.

Each item is one column of

$$\mathsf{M} = \begin{array}{c} \\ \theta \\ b_{1,1} \\ b_{1,2} \\ b_{1,3} \\ b_{2,1} \\ b_{2,2} \\ b_{2,3} \end{array} \begin{array}{c} \begin{array}{cccccc} \theta & \Delta_{2,g} & t_{2,g} & \beta_1 & \beta_2 & \beta_3 \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/D_1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/D_2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/D_3 \\ 0 & 1 & \zeta_1 & s_1/D_1 & 0 & 0 \\ 0 & 1 & \zeta_2 & 0 & s_2/D_2 & 0 \\ 0 & 1 & \zeta_3 & 0 & 0 & s_3/D_3 \end{pmatrix} \end{array}. \tag{9}$$

**The multiplication.** Column $j$ of $\mathsf{D} = \mathsf{BM}$ is the sum of the columns of $\mathsf{B}$, weighted by column $j$ of $\mathsf{M}$:

$$\begin{aligned} \frac{\partial m}{\partial \theta} &= y, & \frac{\partial m}{\partial \Delta_{2,g}} &= v_1 + v_2 + v_3, \\ \frac{\partial m}{\partial t_{2,g}} &= \zeta_1 v_1 + \zeta_2 v_2 + \zeta_3 v_3, & \frac{\partial m}{\partial \beta_i} &= \frac{u_i + s_i v_i}{D_i}. \end{aligned} \tag{10}$$

Written out entry by entry, the offset column is

$$\frac{\partial m}{\partial \Delta_{2,g}} = \left( \frac{\partial C^{11}}{\partial b_{2,1}}, \frac{\partial C^{22}}{\partial b_{2,2}}, \frac{\partial C^{33}}{\partial b_{2,3}}, \frac{\partial C^{12}}{\partial b_{2,1}} + \frac{\partial C^{12}}{\partial b_{2,2}}, \frac{\partial C^{23}}{\partial b_{2,2}} + \frac{\partial C^{23}}{\partial b_{2,3}} \right)^{\mathsf{T}}: \tag{11}$$

each auto spectrum responds to its own bin's $b_{2,i}$, and each cross spectrum to both of its bins'.

**Another model, the same base.** The free-per-bin model replaces the group offset and tilt by one offset per bin, $b_{2,i} = b_2^{\mathrm{L}}(b_{1,i}) + \Delta_{2,i}$. Each offset moves one knob, so its block on the $b_{2,i}$ rows is the $3 \times 3$ identity:

$$\mathsf{M}_{\mathrm{free}} = \begin{array}{c} \\ \theta \\ b_{1,1} \\ b_{1,2} \\ b_{1,3} \\ b_{2,1} \\ b_{2,2} \\ b_{2,3} \end{array} \begin{array}{c} \begin{array}{ccccccc} \theta & \Delta_{2,1} & \Delta_{2,2} & \Delta_{2,3} & \beta_1 & \beta_2 & \beta_3 \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/D_1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/D_2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1/D_3 \\ 0 & 1 & 0 & 0 & s_1/D_1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & s_2/D_2 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & s_3/D_3 \end{pmatrix} \end{array}, \qquad \frac{\partial m}{\partial \Delta_{2,i}} = v_i. \tag{12}$$

**The priors.**

- Group model: $1/w_g^2$ on $\Delta_{2,g}$ (a Gaussian prior of width $w_g$), nothing on $t_{2,g}$ (a flat prior).
- Free per bin: $1/w_g^2$ on each $\Delta_{2,i}$.
- Free per bin, $\sqrt{N}$-scaled: the same $\mathsf{M}_{\text{free}}$, with $1/(3\,w_g^2)$ on each $\Delta_{2,i}$. The mean of the three offsets then has the width $w_g\sqrt{3}/\sqrt{3} = w_g$ of the group offset.

Three models, one $\mathsf{B}$: the group and free models differ in $\mathsf{M}$, the free model and its $\sqrt{N}$-scaled version only in $F_{\text{prior}}$. The base, the expensive part, never changes.

### 5.3 Every notebook model as a matrix on the base

The toy's recipes carry over to the paper model, with six bias families instead of two and six groups (BGS, LRG1, LRG2, LRG3, ELG1, ELG2) instead of one. The model is one matrix $\mathsf{M}$ of Eq. (4) on $\mathsf{B}$, written down without any evaluation (`variant_columns` in the notebook). Figure 1 draws it. Its columns, in the figure's order:

- **Cosmology and $\sigma_v$** ($\theta_k$, $\sigma_{v,k}$): unit vectors, as $\theta$ in the toy.
- **The group offsets and tilts** ($\Delta_{2,g}$, $t_{2,g}$, $\Delta_{s,g}$, $t_{s,g}$, $\alpha_{0,g}$ and its tilt, drawn for one group $g$; the ellipsis stands for the other groups and for $\mathrm{SN}_{0,g}$ and $\mathrm{SN}_{2,g}$, which never tilt). An offset puts ones on the rows of the group's bins in its family's block, as $\Delta_{2,g}$ in the toy; a tilt puts the weights $\bar{z}_i - \bar{z}_g$ on the same rows, as $t_{2,g}$, in every group of more than one bin, with a flat prior.
- **The per-bin amplitudes** ($\beta_1, \ldots, \beta_N$): the linear bias of each bin, $\beta_i = b_{1,i}\,D(\bar{z}_i)$. The paper model parameterizes $b_{1,i}$ through the group amplitude $B_g$, $b_{1,i} = B_g/D(\bar{z}_i)$, a free $b_1$ tilt per group, and the correlated residuals $\epsilon_i$ of the pessimistic prior; all three move the bins through these same columns, and the prior on the residuals enters $F_{\text{prior}}$, not $\mathsf{M}$ (Section 7). Through the calibrations of Section 4, $b_{2,i}$ follows $b_{1,i}$ with the slope $b_2^{\mathrm{L}\prime}$ and $b_{s^2,i}$ with $-4/7$, so the column has three entries:

$$\frac{\partial m}{\partial \beta_i} \;=\; \frac{1}{D(\bar{z}_i)}\left[\frac{\partial m}{\partial b_{1,i}} + b_2^{\mathrm{L}\prime}(b_{1,i})\,\frac{\partial m}{\partial b_{2,i}} - \frac{4}{7}\,\frac{\partial m}{\partial b_{s^2,i}}\right]. \tag{13}$$

Other models are other matrices $\mathsf{M}$ on the same base, built in the exploration notebooks (the correlated-prior note); switching needs no evaluation of the data vector.

On $\Delta d_{\text{com}} = 62$ Mpc the paper model's $\mathsf{M}$ has $n_{\text{par}} = 14 + 5\times 6 + 3\times 6 + 76 = 138$ columns: the cosmology and $\sigma_v$, five per-group families, the three tilt families, and one amplitude per bin.

## 6 Computing the base

The base columns are computed once per layout and cached on disk, by the notebook's section 4 or by the MPI script. This section is the machinery: the finite-difference stencil, the batched per-bin columns, the likelihood subclasses through which the steps reach CosmoLike, the exact columns of the linear terms, and what it all costs.

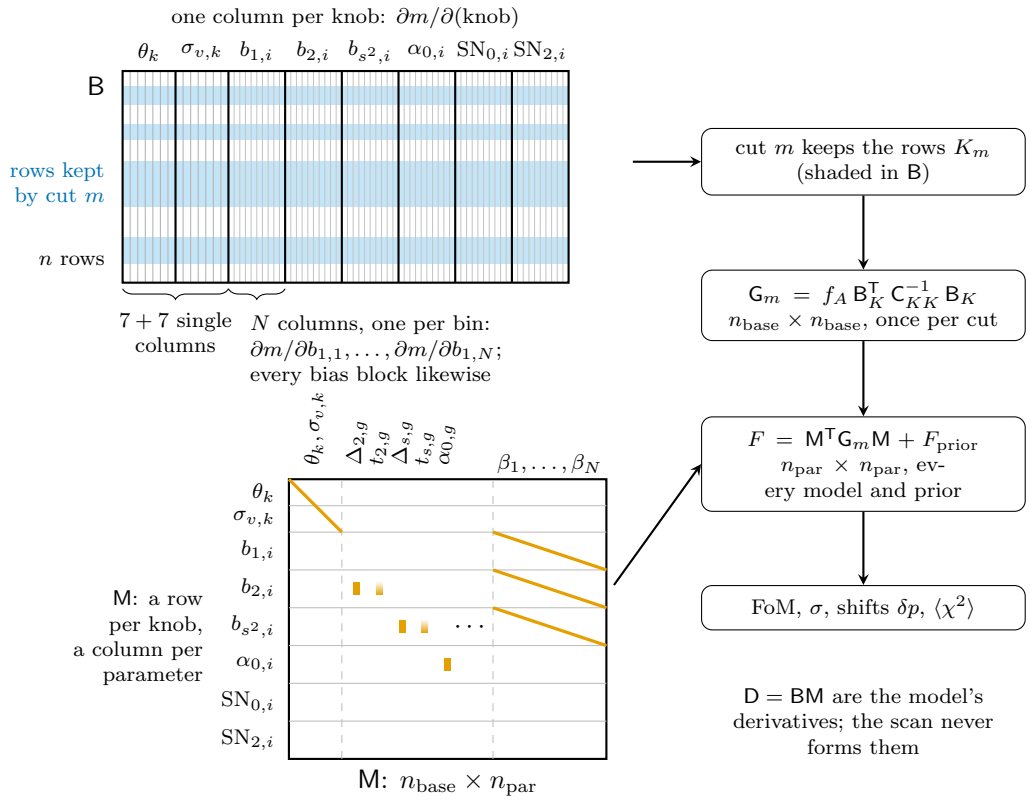



Figure 1: How the notebook assembles a Fisher matrix. Top left: the base $\mathsf{B}$ of one layout. Each cosmological parameter $\theta_k$ and each velocity dispersion $\sigma_{v,k}$ has one column; every bias family is a block of $N$ columns, one per thin bin (thin gray lines). Shaded: the rows a cut $m$ keeps. Bottom left: a model's recipe matrix $\mathsf{M}$, with a row per knob, in the column order of $\mathsf{B}$, and a column per model parameter. Orange marks its nonzero entries: the identity on the cosmology and $\sigma_v$; for one group $g$, the offset $\Delta_{2,g}$ (ones on the group's bins in the $b_{2,i}$ block), its tilt $t_{2,g}$ (the weights $\bar{z}_i - \bar{z}_g$, drawn as a gradient), and likewise $\Delta_{s,g}$, $t_{s,g}$ and $\alpha_{0,g}$, the ellipsis standing for the other groups and the stochastic amplitudes; and three diagonals for the per-bin amplitudes $\beta_i$, Eq. (13). Right: a cut is reduced once to $\mathsf{G}_m$, and every model and prior at that cut is $\mathsf{M}^{\mathsf{T}}\mathsf{G}_m\mathsf{M}$ plus priors; $f_A = 12{,}300/14{,}000$ rescales the inverse covariance to the survey area (Section 3). The $\alpha_{2,i}$ block and the check column of $\mathsf{B}$ are not drawn.

### 6.1 How a derivative column is computed

**A finite difference.** The code cannot differentiate CAMB and CosmoLike by hand. It moves one knob by a small step $\epsilon$, recomputes the data vector, and divides the change by the step. The simplest such finite difference, the central difference $[m(\theta_0 + \epsilon) - m(\theta_0 - \epsilon)]/(2\epsilon)$, is off by a term of order $\epsilon^2$ times the third derivative of $m$: the curvature it ignores.

**The five-point stencil.** A stencil is the set of points at which $m$ is evaluated, with the weights that combine them. The code uses five points, $\theta_0 \pm \epsilon$, $\theta_0 \pm 2\epsilon$ and $\theta_0$, with the weights

$$\frac{\partial m}{\partial \theta} \simeq \frac{m(\theta_0 - 2\epsilon) - 8\, m(\theta_0 - \epsilon) + 8\, m(\theta_0 + \epsilon) - m(\theta_0 + 2\epsilon)}{12\,\epsilon}. \tag{14}$$

These weights cancel the error terms through fourth order, so the remaining error falls as $\epsilon^4$, and the formula is exact for any dependence up to quartic. The center point has weight zero and is never evaluated, so one column costs four evaluations.

**The cosmological and $\sigma_v$ columns.** These are stencils of the sampled parameters of the likelihood, taken by `model.derivatives` of the Fisher module (`interface/cosmolike_lssty1xdesi_fisher.py`). The step is relative, $\epsilon = h\,\theta_0$ with $h = 0.02$; a parameter whose fiducial value is zero and whose prior is flat ($w_a$, the one such case here) takes the absolute step $h$. No step depends on a prior width, which is why changing a prior never invalidates a cached column. Each evaluation is a full call of the likelihood, CAMB included, with the cache of previous results of cobaya (the framework that runs the likelihood) disabled, so that cobaya never hands back a vector saved at an earlier call in place of a fresh one (Appendix A).

The same call also returns the stencils of the six group amplitudes $B_g$, which set the linear bias of every bin of group $g$ at the fiducial point, $b_{1,i} = B_g/D(\bar{z}_i)$. They enter no Fisher matrix of the notebooks, because the per-bin columns replace them; they are kept for the chain-rule check of Section 10.

### 6.2 The per-bin columns, in batches

None of the four families $\{b_{1,i}\}$, $\{b_{2,i}\}$, $\{b_{s^2,i}\}$ and $\{\alpha_{0,i}\}$ of the galaxy power spectrum, Eq. (3), enters $m$ linearly (Section 6.3), so each needs a stencil.

**The stencil of one bin.** For the linear bias of bin $i$, with every other knob at its fiducial value:

1. evaluate $m$ four times, with $b_{1,i}$ replaced by $b_{1,i}(1 \pm \epsilon)$ and $b_{1,i}(1 \pm 2\epsilon)$;
2. combine the four vectors with the weights of Eq. (14), the step being $\epsilon\, b_{1,i}$:

$$\frac{\partial m}{\partial b_{1,i}} \simeq \frac{m\big(b_{1,i}(1-2\epsilon)\big) - 8\,m\big(b_{1,i}(1-\epsilon)\big) + 8\,m\big(b_{1,i}(1+\epsilon)\big) - m\big(b_{1,i}(1+2\epsilon)\big)}{12\,\epsilon\, b_{1,i}}. \quad (15)$$

The step of $b_{1,i}$ is relative, a fraction $\epsilon = 0.01$ of $b_{1,i}$. The steps of the other three are absolute: the four points are $b_{2,i} \pm \epsilon$ and $b_{2,i} \pm 2\epsilon$ with $\epsilon = 0.02$, the same for $b_{s^2,i}$, and $\epsilon = 5\,h^{-2}\,\mathrm{Mpc}^2$ for $\alpha_{0,i}$.

**The problem: one bin at a time is wasteful.** Done bin by bin, a family costs 4 evaluations per bin: $4 \times 76 = 304$ on $\Delta d_{\rm com} = 62$ Mpc, and 1,216 for the four families. Most of that work is wasted. The data vector of that layout holds 183 spectra, 76 autos and 107 crosses, and bin $i$ enters at most six of them: its own auto spectrum and up to five cross spectra. Every other row of $\partial m/\partial b_{1,i}$ is exactly zero, and an evaluation that moves one bin leaves 177 or more spectra unchanged. Counted per spectrum, the block of one family, 183 spectra by 76 columns, is 98% zeros; in the cached arrays no column has a nonzero entry outside its own bin's spectra.

| spectrum | $\partial/\partial b_{1,1}$ | $\partial/\partial b_{1,2}$ | $\partial/\partial b_{1,3}$ | $\partial/\partial b_{1,4}$ | $\partial/\partial b_{1,5}$ |
|---|---|---|---|---|---|
| $C^{11}$ | A | 0 | 0 | 0 | 0 |
| $C^{22}$ | 0 | B | 0 | 0 | 0 |
| $C^{33}$ | 0 | 0 | A | 0 | 0 |
| $C^{44}$ | 0 | 0 | 0 | B | 0 |
| $C^{55}$ | 0 | 0 | 0 | 0 | A |
| $C^{12}$ | A | B | 0 | 0 | 0 |
| $C^{23}$ | 0 | B | A | 0 | 0 |
| $C^{34}$ | 0 | 0 | A | B | 0 |
| $C^{45}$ | 0 | 0 | 0 | B | A |

Table 3: Five bins in a row, cross spectra between neighbors only. Each entry names the batch whose evaluations fill it: batch A moves bins 1, 3 and 5, batch B moves bins 2 and 4. A 0 is exactly zero, because the spectrum does not contain that bin.

**The idea: move several bins in one evaluation.** Take five bins in a row whose only measured cross spectra join neighbors: 1–2, 2–3, 3–4 and 4–5. Put the bins in two batches, sets of bins moved together in one stencil:

- batch A moves $b_{1,i}$ in bins 1, 3 and 5 together;
- batch B moves $b_{1,i}$ in bins 2 and 4 together.

Each batch is one five-point stencil, four evaluations. Figure 2 shows batches on a real layout: each bin is a dot, a line joins two bins whose cross spectrum is in the data vector, and the number in each dot is the bin's batch. Table 3 shows which batch fills each entry of the five columns $\partial m/\partial b_{1,1}, \ldots, \partial m/\partial b_{1,5}$.

Reading the table:

- An auto spectrum $C^{ii}$ is filled in the batch that moves bin $i$.
- A cross spectrum $C^{ij}$ appears in two columns, filled from two different batches: $\partial C^{ij}/\partial b_{1,i}$ from the batch where bin $i$ moves and bin $j$ stays put, and $\partial C^{ij}/\partial b_{1,j}$ from the batch where $j$ moves and $i$ stays put.
- A 0 is not computed: a spectrum that does not contain bin $i$ cannot depend on $b_{1,i}$, so the code sets those rows to zero.

Bin by bin, the five columns would cost $5 \times 4 = 20$ evaluations; the two batches cost $2 \times 4 = 8$.

**A batch is a set of bins, not of spectra.** No batch holds "the autos" or "the crosses". In the evaluations of the batch that contains bin $i$, bin $i$'s auto spectrum and all of its cross spectra receive their derivative with respect to $b_{1,i}$. A cross spectrum $C^{ij}$ is visited twice, once in the batch of each of its two bins. Figure 3c shows this on a real block.

**When may two bins move together?** Why not move every bin at once and keep, for each column, only its own rows?

- That works for the auto spectra: $C^{ii}$ depends only on bin $i$'s biases.

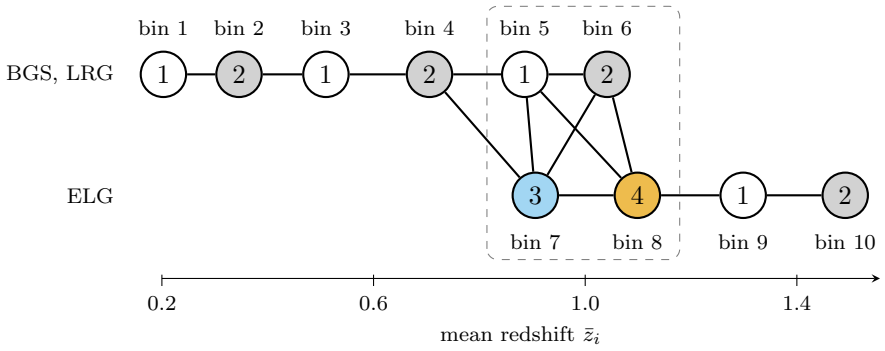



Figure 2: The cross-pair graph of the $\Delta z = 0.2$ layout, all ten bins. Each dot is a bin at its mean redshift (top row BGS and LRG, bottom row ELG), and a line joins two bins whose cross spectrum is in the data vector. The number in each dot is its batch from the greedy rule, repeated by the dot's shade. Bins 5 to 8 (dashed box: two LRG3 and two ELG1 bins, which overlap in redshift) are all joined to each other, so they need four different batches.

- It fails for the cross spectra. If bins $i$ and $j$ move together by $\delta$, $C^{ij}$ changes by $(\partial C^{ij}/\partial b_{1,i} + \partial C^{ij}/\partial b_{1,j})\,\delta$, and one evaluation cannot split the two terms.

So two bins may share a batch only if no measured spectrum contains both. Batching is "move as many bins at once as possible, then keep each column's own rows", with the fewest batches this rule allows.

**What a batch is, as a picture.** Draw each bin as a dot, and join two dots with a line when the data vector contains their cross spectrum: this is the cross-pair graph (Figure 2).

- A batch is a set of dots with no line between any two of them, so its bins can move together.
- Batching the bins means giving every dot a batch number, so that two joined dots never share a number.
- The code (`pair_coloring`) uses a greedy rule: it goes through the bins in data-vector order and gives each the lowest batch number that none of its already numbered neighbors has.

Mathematicians call such batches colors, from the picture of labelling the dots so that joined dots never share a label; hence the name of the code's function.

**The real layout, step by step.** Take $\Delta d_{\rm com} = 62$ Mpc. Its $N = 76$ bins are numbered 1 to 76 in data-vector order, by group and, within a group, by redshift (numbers start at 1, as everywhere in this note and in Figure 3; the code counts from 0). DESI's galaxies come in six groups, ordered by redshift (Table 4). Figure 3 draws the whole layout and its batches; the steps below walk through it.

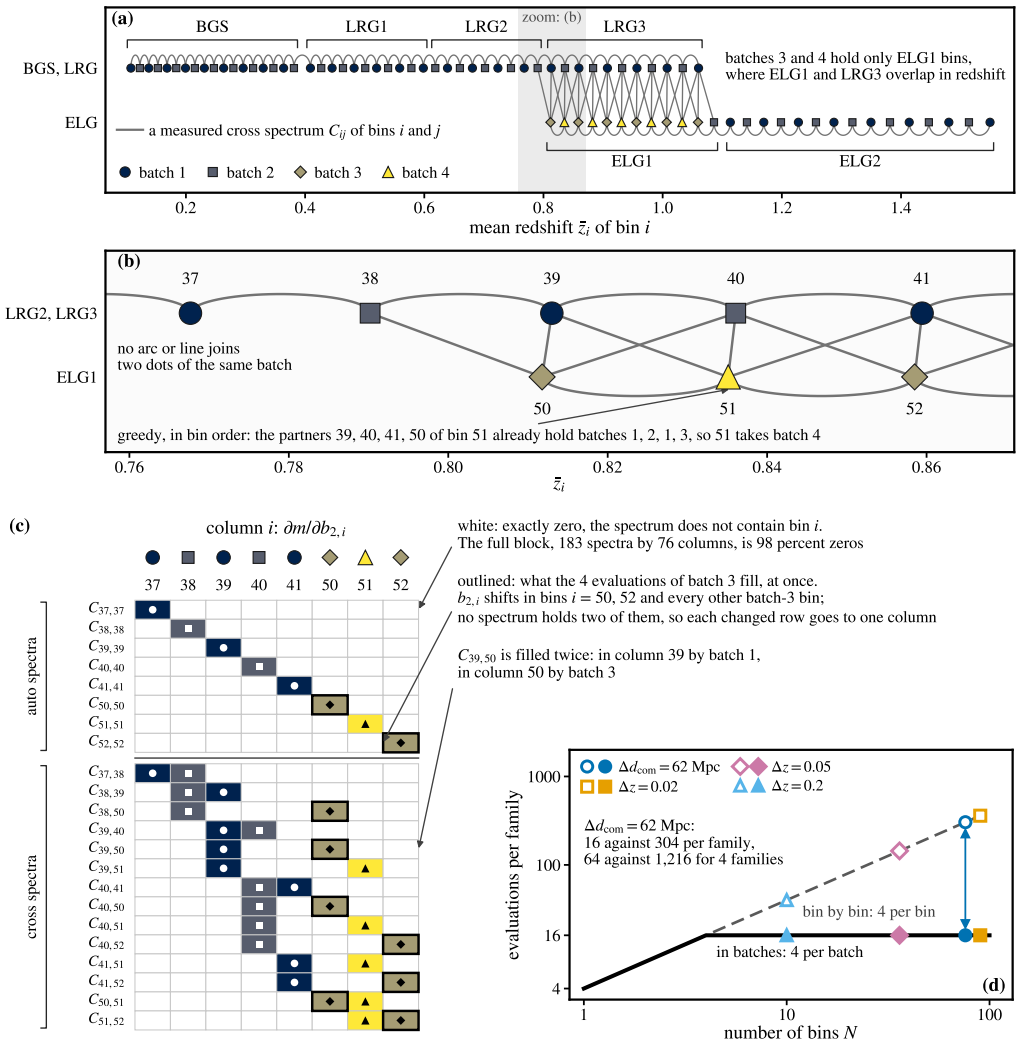



Figure 3: How the per-bin bias columns of $\Delta d_{\rm com} = 62$ Mpc are computed in batches. (a) Every bin at its mean redshift, BGS and LRG on the top row, ELG on the bottom; an arc or line is a measured cross spectrum, and the marker gives the batch. Look first at the long chains: each bin is crossed only with its neighbors, so the markers alternate between batch 1 (circle) and batch 2 (square). Then look at the overlap, where LRG3 and ELG1 cover the same redshifts: the lines between the rows are their cross spectra, and only there do batches 3 (diamond) and 4 (triangle) appear. (b) A zoom on the gray band of (a). No arc or line joins two markers of the same batch. Follow the greedy rule at bin 51: its earlier partners 39, 40, 41 and 50 already hold batches 1, 2, 1 and 3, so it takes batch 4. (c) The block $\partial m/\partial b_{2,i}$ of those eight bins, with a row per spectrum and a column per bin. A column is nonzero only on the spectra that contain its bin; white cells are exactly zero. The outlined cells are what the four evaluations of batch 3 fill at once. A cross spectrum such as $C_{39,50}$ is filled twice: in column 39 by batch 1, and in column 50 by batch 3. (d) Evaluations per bias family against the number of bins $N$: bin by bin the cost grows as $4N$ (open markers, dashed line); in batches it stops at $4 \times 4 = 16$ (filled markers), whatever $N$.

| group | galaxies | bins | mean redshifts $\bar{z}_i$ |
|---|---|---|---|
| BGS | bright galaxies | 1–18 | 0.11–0.38 |
| LRG1 | luminous red galaxies | 19–29 | 0.41–0.60 |
| LRG2 | luminous red galaxies | 30–38 | 0.62–0.79 |
| LRG3 | luminous red galaxies | 39–49 | 0.81–1.06 |
| ELG1 | emission-line galaxies | 50–61 | 0.81–1.09 |
| ELG2 | emission-line galaxies | 62–76 | 1.11–1.55 |

Table 4: The six groups of $\Delta d_{\rm com} = 62$ Mpc. LRG3 and ELG1 cover the same redshifts: they overlap.

**Which cross spectra are measured.** The rule is simple: the data vector holds the cross spectrum of two bins when their redshift distributions $n(z)$ overlap or touch, that is, when the bins are neighbors in redshift. Bins far apart in redshift share no spectrum. For the fixed-$\Delta z$ layouts this is the rule of `cross_pairs` in `scripts/make_desi_dz_case.py`. On $\Delta d_{\rm com} = 62$ Mpc the pair list is built by `scripts/make_desi_pairs.py`: it starts from the 97 pairs the DESI forecast measures and adds every pair whose kernels, blurred by the Fingers-of-God smoothing, still overlap, $O_{ij} \geq 10^{-6}$ with $O_{ij}$ the normalized overlap of the two blurred $n(z)$. That adds ten LRG3–ELG1 pairs, for 107 in all. The threshold is not a tuning: any value between $10^{-6}$ and 0.0277 gives the same lists, and the blur adds no pair on the fixed-$\Delta z$ layouts. The cross-correlation note, `notes/cross_correlations`, derives the rule and the blur.

On $\Delta d_{\rm com} = 62$ Mpc the 107 cross spectra come in two kinds:

1. 74 join a bin to the next one along the chain of bins: neighboring thin shells, which touch at their shared edge. These include the four links at group boundaries: BGS–LRG1, LRG1–LRG2, LRG2–LRG3 and ELG1–ELG2.

2. 33 join an LRG bin to an ELG1 bin at nearly the same redshift: 32 LRG3–ELG1 pairs and one LRG2–ELG1 pair at $z \approx 0.8$. This is the overlap, explained below.

Each bin has 1 to 5 cross partners.

**Along the chain: batches 1 and 2.** The greedy rule walks through the bins in order. Bin 1 takes batch 1. Bin 2 is crossed with bin 1, so it takes batch 2. Bin 3 is crossed with bin 2 only, among the bins already numbered, so it takes batch 1 again, and so on: along the chain the bins alternate.

- The evaluations of batch 1 move bins $1, 3, 5, \ldots$ They fill $\partial C^{11}/\partial b_{1,1}$, $\partial C^{33}/\partial b_{1,3}$, $\ldots$, and the cross spectra $C^{12}$, $C^{23}$, $C^{34}$ through their odd end: $\partial C^{12}/\partial b_{1,1}$, $\partial C^{23}/\partial b_{1,3}$, $\partial C^{34}/\partial b_{1,3}$.

- The evaluations of batch 2 move bins $2, 4, 6, \ldots$ and fill the even ends: $\partial C^{12}/\partial b_{1,2}$, $\partial C^{23}/\partial b_{1,2}$, $\partial C^{34}/\partial b_{1,4}$, and the even auto spectra.

This is the five-bin example of Table 3, continued through every group: the alternating circles and squares of Figure 3a.

**The overlap: why batches 3 and 4 appear.** LRG3 and ELG1 are two different galaxy populations, luminous red galaxies and emission-line galaxies, observed in the same volume, at $z \approx 0.8$–$1.1$ (Table 4). An LRG3 bin and an ELG1 bin at the same redshift trace the same matter, so their cross spectrum is measured, roughly $b_{1,i}\, b_{1,j}\, P$ with $P$ the matter power spectrum. It carries no shot-noise term, because the two bins hold different galaxies: the $\mathrm{SN}_{0,i}$ cross terms exist only within one sample.

These pairs obey the same rule: the two bins are neighbors in redshift. They only look far apart because the bin numbers run group by group. LRG3 bin 44 ($\bar{z} = 0.932$) and ELG1 bin 55 ($\bar{z} = 0.931$) are 11 apart in number but at the same redshift, and their cross spectrum is measured. In Figure 2, drawn against redshift, the lines between the LRG and ELG rows are short.

For batching this matters because an ELG1 bin's redshift neighbors are its ELG neighbors and up to three LRG bins at nearly the same redshift. Bin 51, for example, is crossed with bins 50 and 52 (ELG1) and with bins 39, 40 and 41 (LRG3). Bins 39, 40 and 41 already sit in batches 1, 2 and 1, and bin 50 in batch 3, so bin 51 takes batch 4. The greedy rule puts the eleven ELG1 bins 50 to 60 in batches 3 (bins $50, 52, \ldots, 60$) and 4 (bins $51, 53, \ldots, 59$); batch 1 holds 33 bins and batch 2 holds 32, and the last ELG1 bin, 61, lands in batch 2. Batches 3 and 4 appear only in the overlap (Figure 3b).

**Four batches, whatever the number of bins.** Every bin's partners are local in redshift, so the number of batches does not grow with $N$. The other layouts follow the same pattern, neighbors along the chain plus overlap pairs: $\Delta z = 0.02$ has 133 cross spectra (88 along the chain) and 1 to 5 partners per bin; $\Delta z = 0.05$ has 52 (34) and 1 to 5; $\Delta z = 0.2$ has 13 (8) and 1 to 4. All four layouts get four batches.

A rule of thumb explains why the number stays small: if every bin's partners lie within $k$ places of it in redshift order, $k + 1$ batches suffice, because going through the bins in that order, each bin has at most $k$ numbered partners to avoid. The partners reach 3 places away on every layout. Four batches is also the minimum on every layout: each contains four bins all crossed with each other, two LRG3 and two ELG1 bins at the same redshift (bins 40, 41, 51 and 52 on $\Delta d_{\mathrm{com}} = 62$ Mpc; 36, 37, 51 and 52 on $\Delta z = 0.02$; 15, 16, 21 and 22 on $\Delta z = 0.05$; 5 to 8 on $\Delta z = 0.2$, Figure 2), and four bins all crossed with each other need four different batches.

On every layout batches 3 and 4 hold only ELG1 bins. On $\Delta d_{\mathrm{com}} = 62$ Mpc they are 11 of the 76 bins, about 15%, yet they take half of each family's evaluations.

**What batching buys.** The cost now follows from the number of batches, not from $N$. On $\Delta d_{\mathrm{com}} = 62$ Mpc:

1. one family takes 4 batches $\times$ 4 evaluations = 16 evaluations and fills all 76 of its columns, instead of $4 \times 76 = 304$;

2. the four stenciled families take $4 \times 16 = 64$ evaluations, instead of 1,216.

The same 64 hold on every layout, from 10 bins to 90 (Figure 3d).

**Why not exact derivatives by hand?** If each cross spectrum were a fixed bilinear expression in the biases, its derivatives could be written down exactly. The low-rank non-Limber

engine of CosmoLike does not compute it that way. For each bin it approximates $\sqrt{P_{g,i}(k,z)}$, with $P_{g,i}$ the one-loop galaxy power spectrum of bin $i$, by a short sum of products of a function of $k$ and a function of $z$: a truncated singular-value decomposition, which makes the projection fast.

It builds a cross spectrum from the geometric mean $\sqrt{P_{g,i}}\sqrt{P_{g,j}}$ of the two bins' spectra. Before the square root it floors $P_{g,i}$ at $10^{-15}\, b_{1,i}^2 P_\delta$, with $P_\delta$ the matter power spectrum, so the root always exists. The square root, the truncation and the floor make the spectra not exactly bilinear in the biases. Finite differences through the full likelihood differentiate the model exactly as it is computed.

**How the steps reach CosmoLike.** At every evaluation the likelihood computes all per-bin biases from the sampled parameters and hands them to CosmoLike through compiled setters. No sampled parameter moves one bin alone, so the base is evaluated on a subclass of each paper likelihood (`scripts/compute_fisher_base_mpi_classes.py`) that can add one per-bin step: it first sends the point's own arrays, exactly as its parent does, and then calls the stepped operator's setter once more with the step added to the batch's bins. CosmoLike thus receives a complete, consistent set of values and rebuilds only what depends on the ones that changed. Cobaya's cache of previous results is bypassed (`cached=False`), since every stencil evaluation is at the same sampled point. The code and the details, including what tests this, are in Appendix A.

### 6.3 Exact templates

**Linear terms need no stencil.** Suppose the data vector depends on an amplitude $A$ only through one added term,

$$m(\theta, A) \;=\; m_{\rm rest}(\theta) \;+\; A\,T(\theta). \tag{16}$$

Then $\partial m/\partial A = T(\theta)$ exactly. No step enters, so there is no truncation error at all, not even the $\epsilon^4$ error of the five-point stencil, Eq. (14). $T$, the term per unit amplitude, is called the *template* of $A$: one evaluation of the term, or the term computed directly from the fiducial kernels, gives the exact column. Three families enter $m$ this way: the redshift-space counterterm $\{\alpha_{2,i}\}$ (below, under "Term 2"), and two whose templates the likelihood computes directly,

- the white noise beyond Poisson $\mathrm{SN}_{0,i}$, in units of the bin's Poisson power $1/\bar{n}_i$ ($\bar{n}_i$ its tracer density);
- the stochastic term $\mathrm{SN}_{2,i}\, k^2\mu^2$, with $\mu$ the cosine of the angle to the line of sight, in units of $f_{\rm sat}\sigma_{\rm sat}^2/\bar{n}_i$ ($f_{\rm sat}$ the satellite fraction, $\sigma_{\rm sat}$ the satellites' velocity dispersion in distance units).

That these are *exactly* linear is not obvious: the density–density part of every DESI spectrum goes through a singular value decomposition and a square root ("Why not exact derivatives by hand?"), and a square root is linear in nothing. The resolution is that none of the three families ever enters that route. To see it, write the whole spectrum once, term by term.

**The whole $C_\ell$, term by term.** The pipeline evaluates the cross spectrum of bins $i$ and $j$ as a sum of two terms, each projected by its own route, and the likelihood adds the stochastic

| term | route | depends on | linear in its amplitude? |
|---|---|---|---|
| density–density | low-rank (SVD, floor) | $b_{1,i}$, $b_{2,i}$, $b_{s^2,i}$, $\alpha_{0,i}$; $\theta$ | no: stencils |
| redshift-space rows | FKEM split | $b_{1,i}$; $f$, $D$ | no amplitude of its own |
| $\alpha_{2,i}$ counterterm | the same rows | $\alpha_{2,i}$; $b_{1,i}$, $f$ | yes: a $\pm$ pair is exact |
| $\mathrm{SN}_{0,i}$, $\mathrm{SN}_{2,i}$ | likelihood templates | $\mathrm{SN}_{0,i}$, $\mathrm{SN}_{2,i}$; $W_i$ | yes: computed directly |

Table 5: The terms of a DESI band power, Eqs. (17) and (21): the route each takes, what it depends on, and whether it is exactly linear in an amplitude of its own. $\theta$ stands for the cosmological parameters and the velocity dispersions $\sigma_v$.

terms afterwards:

$$C_\ell^{ij} \;=\; \underbrace{C_\ell^{\mathrm{DD},\,ij}}_{\substack{\text{low-rank route:}\\ \text{SVD, geometric mean, floor}}} \;+\; \underbrace{C_\ell^{\mathrm{RSD},\,ij}}_{\substack{\text{FKEM split:}\\ \text{density}\times\text{RSD, RSD}\times\text{RSD}}} \quad . \tag{17}$$

Every DESI row of the data vector is the average of $C_\ell^{ij}$ over one band of multipoles, and averaging is linear, so every linearity below survives into the band powers. Lensing magnification would be a fourth term; this forecast sets its amplitude to zero in every DESI bin. Table 5 summarizes the terms; the paragraphs below take them one at a time.

**Term 1: density–density, the only term through the SVD.** $C_\ell^{\mathrm{DD}}$ projects the real-space galaxy power $P_{gg,i}$ of Eq. (3). Per bin, the low-rank engine factorizes $\sqrt{P_{g,i} - P_{\infty,i}}$ by a singular value decomposition into five products of a function of $k$ and a function of $z$; here $P_{\infty,i} = -(\sigma_4/2)\, D^4\, (b_{2,i} + 2b_{s^2,i}/3)^2$ is the white-noise constant that the loop subtraction leaves at high $k$ ($\sigma_4$ a fixed integral of the linear power), and a constant power projects exactly in Limber, so it is carried there instead. A cross pair uses the geometric mean $\sqrt{P_{g,i}}\sqrt{P_{g,j}}$, and a floor guards the square root. This term depends on $b_{1,i}$, $b_{2,i}$, $b_{s^2,i}$ and $\alpha_{0,i}$ – and, through the square root, the rank truncation and the floor, on none of them exactly linearly.

**Term 2: the redshift-space rows, through the FKEM split.** With $W_i(\chi)$ the bin's kernel – its redshift distribution as a function of comoving distance $\chi$, after the Fingers-of-God smoothing – $D$ the growth factor, $f$ the growth rate and $j_\ell$ the spherical Bessel function, the bin's linear redshift-space kernel is

$$\Delta_\ell^i(k) \;=\; \int d\chi\, W_i\, D\, \left[\, b_{1,i}\, j_\ell(k\chi) - f\, j_\ell''(k\chi) \,\right] \tag{18}$$

(each factor $f\mu^2$ of the flat-sky power becomes a $-j_\ell''$ in the exact projection; the summary note, `notes/bias_model_summary`, on the non-Limber spectra). Writing $\Delta_\ell^{i,\mathrm{D}}$ for the density part alone, the $b_{1,i}\, j_\ell$ term of $\Delta_\ell^i$, the term is

$$C_\ell^{\mathrm{RSD},\,ij} \;=\; \frac{2}{\pi} \int dk\, k^2\, P_{\mathrm{lin}}(k) \left[\, \Delta_\ell^i\, \Delta_\ell^j - \Delta_\ell^{i,\mathrm{D}}\, \Delta_\ell^{j,\mathrm{D}} \,\right] \;+\; \text{Limber corrections on the same rows,} \tag{19}$$

the density×RSD and RSD×RSD products, computed with the separable linear spectrum by FFTLog and corrected in Limber with the full one (the FKEM split). The term depends on $b_{1,i}$, already covered by its stencil, and on cosmology through $f$ and $D$. It has no amplitude of its own.

One amplitude does ride these rows: the redshift-space counterterm $\alpha_{2,i}$, which in the DESI DR1 convention [2] adds $(b_1 + f\mu^2)\, f\, \alpha_{2,i}\, \mu^2 k^2 P_{\rm lin}$ to the bin's redshift-space power – a fixed term times $\alpha_{2,i}$, so the band powers are exactly linear in it and a single central difference, $[m(+\epsilon) - m(-\epsilon)]/(2\epsilon)$ at $\epsilon = 5\,h^{-2}\,\mathrm{Mpc}^2$ through the full likelihood, is the exact column. The columns are taken in the batches of Section 6.2, two evaluations per batch, and the forward and backward differences check the linearity: they agree to at most $1.6 \times 10^{-10}$ of the column over the four layouts. The paper model fixes $\alpha_{2,i} = 0$; the columns serve model B and the shift tests of Section 8.

**Terms 3 and 4: the stochastic templates, added outside CosmoLike.** $\mathrm{SN}_{0,i}$ adds $\mathrm{SN}_{0,i}/\bar{n}_i$ to $P_{g,i}$, a constant in $k$. For a constant power $S$ the $k$ integral of the exact projection closes by the orthogonality of the Bessel functions, $\frac{2}{\pi}\int dk\, k^2\, j_\ell(k\chi_1)\, j_\ell(k\chi_2) = \delta_{\rm D}(\chi_1 - \chi_2)/\chi_1^2$ ($\delta_{\rm D}$ the Dirac delta), so the spectrum gains, independently of $\ell$,

$$C_\ell^{ij} \;\supset\; S \int d\chi\, \frac{W_i(\chi)\, W_j(\chi)}{\chi^2}, \qquad S = \frac{\mathrm{SN}_{0,i}}{\bar{n}_i} \;\text{ on the auto spectrum.} \tag{20}$$

A cross pair inside one tracer sample takes the mean of its two bins' stochastic powers, and a pair of two different samples takes none: different galaxies share no stochastic term. The integral is fixed by the fiducial kernels, so the band powers are exactly linear in $\mathrm{SN}_{0,i}$, with the integral as the template. The $\mathrm{SN}_{2,i}$ term is $S_2\, k^2\mu^2$ with $S_2 = \mathrm{SN}_{2,i}\, f_{\rm sat}\sigma_{\rm sat}^2/\bar{n}_i$; since $k^2\mu^2 = k_\parallel^2$ is the Fourier image of a second derivative along the line of sight, two integrations by parts move one derivative onto each kernel, and the template is $\int d\chi\, W_i' W_j'/\chi^2$ with $W' = dW/d\chi$ (the higher-order note, `notes/higher_order_bias`, on the stochastic terms in projection). The likelihood computes both templates directly from the fiducial kernels (`stochastic_templates` of `likelihood/desi_cl.py`, orders 0 and 2) and adds amplitude times template to the vector CosmoLike returned, keeping the sum in `last_datavector`. The amplitudes never enter CosmoLike's projection at all, let alone the SVD.

**The consequence.** Collecting the terms, the DESI part of the data vector is

$$m \;=\; m_{\rm DD+RSD}\big(\{b_{1,i}\}, \{b_{2,i}\}, \{b_{s^2,i}\}, \{\alpha_{0,i}\};\, \theta\big) + \sum_i \alpha_{2,i}\, T_i^{(a_2)} + \sum_i \mathrm{SN}_{0,i}\, T_i^{(0)} + \sum_i \mathrm{SN}_{2,i}\, T_i^{(2)}, \tag{21}$$

with $\theta$ the cosmological parameters and the $\sigma_v$. The last three groups have the form of Eq. (16), so their columns are exact – the stochastic templates computed directly, the $\alpha_2$ columns by one central difference each; only the first group needs stencils.

**Why $\alpha_{0,i}$ is not a template.** It looks like one: a counterterm amplitude multiplying $k^2 P_{\rm mm}$ in Eq. (3). But it sits in the real-space density–density power, the one term that goes under the square root, the SVD and the geometric mean, so the projected spectra are not exactly linear in it; the same holds for $b_{1,i}$, $b_{2,i}$ and $b_{s^2,i}$. That is the dividing line of this section: the families that stay outside the density–density term get exact templates, and everything that passes through the SVD gets the stencils of Section 6.2.

The templates themselves are validated in the higher-order note, `notes/higher_order_bias`, section "Validation": the exact $\mathrm{SN}_{2,i}$ projection against a brute-force $j_\ell$, $j_\ell''$ double integral, and the linearity of the $\mathrm{SN}_{2,i}$ term by forward against backward differences.

| term | what it is | evaluations |
|---|---|---|
| 1 | the fiducial vector, templates and biases | 1 |
| $4\,N_{\rm batch}\,N_{\rm fam}$ | 4 batches, 4 stenciled families $\{b_{1,i}\}$, $\{b_{2,i}\}$, $\{b_{s^2,i}\}$, $\{\alpha_{0,i}\}$ | 64 |
| $4\,N_{\rm stencil}$ | 20 stencil parameters: 7 cosmology, 7 $\sigma_v$, 6 $B_g$ | 80 |
| $2\,N_{\rm batch}$ | the exact $\{\alpha_{2,i}\}$ columns: 4 batches, + and − | 8 |
| 1 | the fiducial point again, after every step (the reset check) | 1 |
| | total | 154 |

Table 6: The evaluations behind one layout's base, in the order they run. The $\{\mathrm{SN}_{0,i}\}$ and $\{\mathrm{SN}_{2,i}\}$ templates cost no evaluation beyond the fiducial one, whose kernels they are read from.

| configuration | grid factor $g$ | seconds per evaluation |
|---|---|---|
| MCMC settings | 32 | 3.9 |
| starting configuration, grid lowered | 64 | 11.4 |
| starting configuration | 128 | 17.0 |
| starting configuration, grid raised | 256 (final) | 28.0 |
| starting configuration, grid raised | 512 | 48.4 |
| final settings | 256 (final) | 19.5 |

Table 7: The cost of one data-vector evaluation on $\Delta z = 0.05$, measured by the convergence study on a MacBook Pro with the Apple M2 Pro chip (8 OpenMP threads, one BLAS thread), on AC power with no other heavy job running, except the $g = 256$ run of the starting configuration, which started while one other job was running. The final settings ran 130 evaluations in 42 minutes. Section 9 describes the configurations.

### 6.4 How many data-vector evaluations, and how long

Counted in the notebook and the Fisher module, the base of one layout takes

$$N_{\rm eval} \;=\; 1 + 4\,N_{\rm batch}\,N_{\rm fam} + 4\,N_{\rm stencil} + 2\,N_{\rm batch} + 1 \tag{22}$$

evaluations of the data vector, term by term in Table 6. It is 154 on every layout, because no term grows with the number of bins.

**The cost of one evaluation.** Each evaluation is a complete data vector: CAMB, then CosmoLike's non-Limber projection of every thin bin. Its cost depends mostly on $g$, the refinement factor of the comoving-distance grid on which the non-Limber projection integrates (Section 9). Table 7 gives the times the convergence study measured on $\Delta z = 0.05$ (36 bins), on AC power. The MCMC settings are those of the example yamls used for Markov-chain Monte Carlo runs, the sampling of the full likelihood.

**The total time.** The base of a layout takes $N_{\rm eval} \times t_{\rm eval}$, with $t_{\rm eval}$ the time of one evaluation, measured on a MacBook Pro with the Apple M2 Pro chip (8 OpenMP threads, one BLAS thread, on AC power with no other heavy job). At the final settings:

- $\Delta z = 0.05$, from the measured $t_{\rm eval}$: $154 \times 19.5$ s $\approx 50$ minutes.

- The other layouts, a rough estimate that scales $t_{\rm eval}$ with the number of bins: $\Delta d_{\rm com} = 62$ Mpc (76 bins) $154 \times 41$ s $\approx 1.8$ hours; $\Delta z = 0.02$ (90 bins) $154 \times 49$ s $\approx 2.1$ hours; $\Delta z = 0.2$ (10 bins) $154 \times 5.4$ s $\approx 14$ minutes.

The four layouts then take about five hours. On other hardware – an Intel workstation, an x86 cluster node – the time per evaluation differs, so these numbers are a guide to the order of magnitude, not a benchmark.

**The same base on MPI ranks.** `scripts/compute_fisher_base_mpi.py` splits the 154 evaluations of one layout over $N$ MPI ranks, each running CosmoLike's and CAMB's OpenMP loops on its own threads, and writes the same two caches. The evaluations are independent, so rank 0 assembles the gathered vectors with the serial arithmetic; on one rank the script evaluates in the notebook's own order, and on $\Delta z = 0.2$ the $1 \times 8$ and $2 \times 4$ (ranks $\times$ threads) runs reproduce the serial caches bit for bit. Each rank holds a complete model, covariance included, and peaks at about 10 to 15 GB of memory on $\Delta d_{\rm com} = 62$ Mpc, which sets how many ranks a node fits. The project README, section "Computing the derivative base with MPI", gives the launch commands; the script's docstring holds the cache rules and the parallel layout.

# 7 A scale cut is a set of rows

A scale cut removes the band powers that small, nonlinear scales dominate. A cut at $k_{\rm max}$, quoted at $z = 0$, cuts bin $i$ at $k_{\rm max}/D(\bar{z}_i)$, by the non-Limber ultraviolet-fraction criterion of Saraivanov et al. [3] within the shipped mask: a band is kept while less than 5% of it comes from the power spectrum above that wavenumber. The Fisher matrix at a cut follows in four steps.

1. The cut keeps a set $K_m$ of rows of $m$. It drops the other rows of $\mathsf{B}$ and keeps every column: a cut never changes a derivative. $\mathsf{B}_K$ is the kept rows of $\mathsf{B}$, and $\mathsf{C}_{KK}$ the kept block of the covariance.

2. The data part of the Fisher matrix of a model at this cut is $F_{\rm data} = \mathsf{D}_K^{\mathsf{T}}\, f_A\, \mathsf{C}_{KK}^{-1}\, \mathsf{D}_K$, with $f_A = 12{,}300/14{,}000$ the factor that rescales the inverse covariance to the survey area (Section 3). Since $\mathsf{D}_K = \mathsf{B}_K \mathsf{M}$,

$$F_{\rm data} \;=\; \mathsf{M}^{\mathsf{T}} \big( f_A\, \mathsf{B}_K^{\mathsf{T}}\, \mathsf{C}_{KK}^{-1}\, \mathsf{B}_K \big) \mathsf{M} \;=\; \mathsf{M}^{\mathsf{T}}\, \mathsf{G}_m\, \mathsf{M}, \qquad \mathsf{G}_m \;\equiv\; f_A\, \mathsf{B}_K^{\mathsf{T}}\, \mathsf{C}_{KK}^{-1}\, \mathsf{B}_K. \tag{23}$$

   $\mathsf{G}_m$ is what the data of cut $m$ say about every pair of knobs: its entry $(a, b)$ is the information shared by knobs $a$ and $b$.

3. $\mathsf{G}_m$ contains neither $\mathsf{M}$ nor a prior. It depends only on the base, the cut and the covariance, so the notebook computes it once per cut, and every model and prior at that cut reuses it.

4. The priors add their precision: $F = \mathsf{M}^{\mathsf{T}} \mathsf{G}_m \mathsf{M} + F_{\rm prior}$.

**The toy, cut.** Suppose cut $m$ drops the cross spectrum $C^{23}$, which fails the ultraviolet criterion. Then:

- $K = \{C^{11}, C^{22}, C^{33}, C^{12}\}$, and $\mathsf{B}_K$ is the $4 \times 7$ matrix of the first four rows of Eq. (7);
- $\mathsf{C}_{KK}$ is $4 \times 4$, and $\mathsf{G}_m$ is $7 \times 7$;
- the group model's $F$ is $(6 \times 7)(7 \times 7)(7 \times 6) = 6 \times 6$, and the free model's $7 \times 7$, both from the same $\mathsf{G}_m$.

After this cut, $v_3$ survives only on $C^{33}$.

**The real sizes.** On $\Delta d_{\mathrm{com}} = 62$ Mpc, $\mathsf{B}$ is $10{,}494 \times 547$. The shipped mask keeps 5,674 rows, so $\mathsf{B}_K$ is $5{,}674 \times 547$, $\mathsf{C}_{KK}$ is $5{,}674 \times 5{,}674$ and $\mathsf{G}_m$ is $547 \times 547$. $F$ is $138 \times 138$ for the paper model. The large block is handled once per cut, and every Fisher system at that cut (one Fisher matrix with its inverse and results, for each model and prior the notebook evaluates) reuses $\mathsf{G}_m$.

**How $\mathsf{G}_m$ is computed.** The notebook computes $\mathsf{G}_m$ in `layout_grams`, without ever inverting the covariance:

1. it writes $\mathsf{C}_{KK} = LL^{\mathsf{T}}$ with $L$ lower triangular, the Cholesky factorization, a matrix analogue of the square root;
2. it solves $L\,W = \mathsf{B}_K$ for $W$ row by row, which a triangular $L$ allows (forward substitution);
3. it forms $\mathsf{G}_m = f_A\, W^{\mathsf{T}} W$, with $f_A$ the area rescaling of the inverse covariance.

This saves work, avoids amplifying round-off when the covariance is ill-conditioned (when its inverse magnifies small errors), and makes $\mathsf{G}_m$ symmetric with no negative information, by construction.

Each layout gets eight such matrices: the seven cuts $k_{\max} = 0.05, 0.075, \ldots, 0.2\,h/\mathrm{Mpc}$, and the shipped mask alone, the comparison point of the information test below.

**The priors.** $F_{\mathrm{prior}}$ adds precision and needs no data. The diagonal Gaussian priors, all centered on zero, sit on $\sigma_v$ and on the model's group parameters (Section 2): the $\sigma_v$ widths are the exploration fits' for model B and model B + tilts, and six times them for the paper model, the paper yaml's own widths. The counterterm widths, $0.5/k_{\mathrm{max}}^2$, change with the cut, so the diagonal is rebuilt for each cut (`prior_diagonal`).

The linear-bias prior acts on the block of the $\beta_i$. The baseline and pessimistic priors add, group by group, the penalty of their memory model on bin-to-bin wiggles of the linear biases (`group_bias_precision`; the correlated-prior note, `notes/correlated_bias_prior_v2`, on what the prior charges and how it enters the Fisher matrix). The free prior adds nothing, and the growth-tied prior is a change of $\mathsf{M}$.

**Parameters without data.** At small cuts some bins keep no band, and their columns vanish on $K_m$. A parameter whose information at the cut, data plus prior, falls below $10^{-10}$ of its information on the shipped mask with the same priors leaves the fit: it is removed from $\mathsf{M}$ and from $F$, and the quoted FoMs, errors and shifts are over the remaining parameters. A Gaussian prior keeps a parameter in. A cut that keeps no band (as $\Delta z = 0.02$ at $k_{\max} = 0.05\,h/\mathrm{Mpc}$) or leaves a cosmological parameter without data gives no fit.

**The pseudo-inverse.** A plain inverse fails in one more case. On $\Delta d_{\mathrm{com}} = 62$ Mpc at $k_{\max} = 0.075\,h/\mathrm{Mpc}$ only 7 of the 18 BGS bins keep a band. A flat-prior BGS tilt can then lose every band that carries information on it, leaving one combination of parameters with no information at all. Along such a direction a plain inverse amplifies round-off by about $10^{16}$ and can turn it into a spurious shift of sigmas.

`stable_inverse` therefore takes a pseudo-inverse, an inverse that skips the directions with no information:

1. it scales $F$ to unit diagonal, $\tilde{F} = SFS$ with $S = \mathrm{diag}(F)^{-1/2}$, so that the parameters' units do not matter;
2. it diagonalizes $\tilde{F} = V\Lambda V^{\mathsf{T}}$: the columns of $V$ are the parameter combinations measured independently, and the eigenvalues in $\Lambda$ the information in each;
3. it drops the directions whose eigenvalue is below $10^{-12}$ of the largest, and returns

$$F^{-1} \;=\; S\,V_\kappa\,\Lambda_\kappa^{-1}\,V_\kappa^{\mathsf{T}}\,S, \tag{24}$$

   with $\kappa$ the kept directions.

A dropped direction must leave $(w_0, w_a)$, $\Omega_m$ and $A_s$ alone: the scan gives no fit when one touches them by more than $10^{-6}$. The case above, the flat BGS tilts on $\Delta d_{\mathrm{com}} = 62$ Mpc at $k_{\max} = 0.075\,h/\mathrm{Mpc}$, is the mechanism. Across the notebook's $k_{\max}$ scan, 11 of its 168 Fisher systems drop a direction, each touching the scored parameters by at most $3.3 \times 10^{-10}$. The untied comparison of the summary note drops more. Its version with correlated residuals loses the residual trends that duplicate the flat group tilts: 18 directions at most cuts on the three thinner layouts, and up to 33 at the smallest cuts, each touching the scored parameters by at most $1.1 \times 10^{-7}$.

## 8 Everything downstream reuses $\mathsf{G}$

Once $\mathsf{G}_m$ is built, nothing in the notebook touches an $n \times n$ matrix or the data vector again. $F$ is Eq. (23) plus the priors, inverted by Eq. (24), and $\sigma_j$ and the FoMs follow as in Section 3.

**Shifts.** A truth is an assumed true universe that departs from the fitted model, for example one with the redshift-space counterterm $\alpha_{2,i} \neq 0$ when the fit sets $\alpha_{2,i} = 0$ (Section 4). It moves the data by $\delta m$. When $\delta m$ lies in the span of the base columns, $\delta m = \mathsf{B}c$, with $c$ a vector of coefficients:

- $\alpha_{2,i} = 12.5\,h^{-2}\,\mathrm{Mpc}^2$ in every bin $i$ multiplies the exact per-bin $\alpha_2$ columns of the base (Section 6.3) by 12.5;

- an offset truth puts its values on the $b_{2,i}$ and $b_{s^2,i}$ blocks;
- a fractional drift $\delta_i$ of the linear biases is $c = \sum_i \beta_i \, \delta_i \, \mathsf{M}_{\beta_i}$, with $\mathsf{M}_{\beta_i}$ the column of Eq. (13).

To first order the best fit then moves by

$$\delta p \;=\; F^{-1} \mathsf{D}^{\mathsf{T}} \mathsf{C}^{-1} \delta m \;=\; F^{-1} \mathsf{M}^{\mathsf{T}} \, \mathsf{G}_m \, c \tag{25}$$

(the correlated-prior note, on the shift formula). $\delta m = \mathsf{B} c$ is exact for the $\alpha_{2,i}$, $\mathrm{SN}_{0,i}$ and $\mathrm{SN}_{2,i}$ truths, whose terms are linear, and first order for truths in $b_{1,i}$, $b_{2,i}$ and $b_{s^2,i}$, since the band powers are quadratic in them. A truth costs one product with $\mathsf{G}_m$.

**Random truths.** Eq. (25) is linear in $c$, so a family of random truths collapses into one matrix: for drifts $\delta_i$ of the linear biases the $(w_0, w_a)$ shift is $\delta w = A \, \delta$, with $A$ the $(w_0, w_a)$ rows of Eq. (25) applied to the $N$ columns $\beta_i \mathsf{M}_{\beta_i}$. Two thousand truths cost one matrix product.

**The exact $\langle \chi^2 \rangle$.** Whether a fit's error bars are honest is tested by averaging $\chi^2_w = (\hat{w} - w_{\mathrm{true}})^{\mathsf{T}} \mathsf{C}_w^{-1} (\hat{w} - w_{\mathrm{true}})$, with $\hat{w}$ the fitted $(w_0, w_a)$, over the truths and the data noise. Every source is Gaussian and enters linearly, so the average is exact in closed form (the correlated-prior note, on the test statistic):

$$\langle \chi^2_w \rangle \;=\; \Delta^2_{\mathrm{drift}} \;+\; \mathrm{tr}\Big[ \mathsf{C}_w^{-1} \Big( A \, \Sigma \, A^{\mathsf{T}} + R \, \Sigma_\theta \, R^{\mathsf{T}} + N_w \Big) \Big] . \tag{26}$$

- $\Delta_{\mathrm{drift}}$ is the $(w_0, w_a)$ shift, in units of the ellipse, of the known drift alone: the smooth change of the linear biases with redshift that physics predicts.
- $\Sigma$ is the covariance of the truth's remaining scatter of the linear biases.
- $R$ and $\Sigma_\theta$ are the response to, and the prior variance of, any other parameter drawn at random in the accuracy test.
- $N_w$ is the $(w_0, w_a)$ block of $F^{-1} F_0 F^{-1}$, the share of the data noise and the Gaussian priors, with $F_0$ the Fisher matrix without the linear-bias prior.

The notebook reports $\sqrt{\langle \chi^2 \rangle / n_{\mathrm{dof}}}$, with $n_{\mathrm{dof}} = 2$ for $(w_0, w_a)$, and $n_{\mathrm{dof}} = 1$ for $\Omega_m$ or $A_s$ with that parameter's own row; 1 means honest error bars. The random draws only make the percentile bars of the correlated-prior note's figures.

**The cost.** Each $\mathsf{G}_m$ costs one Cholesky factorization of at most 7,621 rows, about $n_K^3/3 \approx 1.5 \times 10^{11}$ floating-point operations, and one triangular solve, eight times per layout. A Fisher system then costs $\mathsf{M}^{\mathsf{T}} \mathsf{G}_m \mathsf{M}$, with $n_{\mathrm{base}} \leq 645$ and $n_{\mathrm{par}} \leq 164$ (model B + tilts on $\Delta z = 0.02$), about $2 \, n_{\mathrm{base}}^2 n_{\mathrm{par}} \approx 1.4 \times 10^8$ operations, plus the diagonalization of an $n_{\mathrm{par}} \times n_{\mathrm{par}}$ matrix: a fraction of a second.

The $k_{\mathrm{max}}$ scan of the notebook has 168 systems (four layouts, seven cuts, model B and model B + tilts, and the baseline, pessimistic and free priors; the growth-tied prior fails the accuracy test and is left out), 162 with a fit and 6 without any band. In operations, all of them together cost less than the Cholesky factorizations of one layout. Without the base, each variant would need its own derivatives, at the cost of Section 6.4 per layout, and each truth its own data vector.

| array | shape | what it is |
|---|---|---|
| `EXAMPLE_EVALUATE_FISHER_MODEL_DERIVATIVES_1.pkl` (33.7 MB) | | |
| `m0` | 10,494 | the fiducial band powers $m(p_0)$ |
| `D` | $10{,}494 \times 20$ | stencil columns: 7 cosmology, 6 $B_g$ (check only), 7 $\sigma_v$ |
| `ops['b1']` | $10{,}494 \times 76$ | column $i$: $\partial m/\partial b_{1,i}$, the linear bias of bin $i$ alone |
| `ops['b2']` | $10{,}494 \times 76$ | column $i$: $\partial m/\partial b_{2,i}$ |
| `ops['bs2']` | $10{,}494 \times 76$ | column $i$: $\partial m/\partial b_{s^2,i}$ |
| `ops['a0']` | $10{,}494 \times 76$ | column $i$: $\partial m/\partial \alpha_{0,i}$, per $h^{-2}\,\mathrm{Mpc}^2$ |
| `T` | $10{,}494 \times 76$ | column $i$: $\partial m/\partial \mathrm{SN}_{0,i}$ |
| `b1`, `b2`, `bs2` | 76 each | the fiducial $b_{1,i}$, $b_{2,i}$, $b_{s^2,i}$ of every bin |
| `names`, `fiducial` | 20 each | the stencil parameters and their fiducial values |
| `priors`, `leaders` | | the $\sigma_v$ prior widths; each bin's group amplitude |
| `EXAMPLE_EVALUATE_FISHER_MODEL_TEMPLATES_1.pkl` (19.2 MB) | | |
| `m0` | 10,494 | the fiducial vector again (the cross-cache check) |
| `a2` | $10{,}494 \times 76$ | column $i$: $\partial m/\partial \alpha_{2,i}$, per $h^{-2}\,\mathrm{Mpc}^2$, exact |
| `T0` | $10{,}494 \times 76$ | the $\mathrm{SN}_{0,i}$ templates again (equal to `T`) |
| `T2` | $10{,}494 \times 76$ | column $i$: $\partial m/\partial \mathrm{SN}_{2,i}$ |
| `nonlinearity`, `reset` | | the $\alpha_2$ forward-backward check; $\max\lvert m_{\mathrm{after}} - m_0\rvert / \max\lvert m_0\rvert$ |

Table 8: The precomputed arrays of $\Delta d_{\mathrm{com}} = 62$ Mpc, in Python's pickle files: $n = 10{,}494$ band powers and $N = 76$ bins. Each $10{,}494 \times 76$ block is 6.4 MB in double precision. The files also hold the settings, tier and fingerprint they were made with (Section 9). The other layouts' files are 47.5/27.3, 8.8/4.8 and 1.1/0.5 MB ($\Delta z = 0.02$, 0.05, 0.2; derivatives/templates), and the scale-cut cache is 6.2 MB. In the toy's notation, `ops['b1']` is $[u_1, \ldots, u_N]$ and `ops['b2']` is $[v_1, \ldots, v_N]$.

### 8.1 What is precomputed, and what is computed at every run

The work splits into three levels.

1. **On disk, once per layout (hours):** the base columns and the fiducial vector $m(p_0)$. Table 8 lists every array on $\Delta d_{\mathrm{com}} = 62$ Mpc.

2. **In memory, at every run (seconds):** for each scale cut $m$, its kept rows $K_m$ and the matrix $\mathsf{G}_m = f_A\, \mathsf{B}_K^{\mathsf{T}} \mathsf{C}_{KK}^{-1} \mathsf{B}_K$, with $f_A = 12{,}300/14{,}000$ the factor that rescales the inverse covariance to the survey area (Section 3), $547 \times 547$ on $\Delta d_{\mathrm{com}} = 62$ Mpc (Section 7). The covariance is read fresh from its file at every run.

3. **For every model and prior (a fraction of a second):** $\mathsf{M}$, then $F = \mathsf{M}^{\mathsf{T}} \mathsf{G}_m \mathsf{M} + F_{\mathrm{prior}}$, its inverse, the FoMs, the shifts and $\langle \chi^2 \rangle$ (Section 7 and this section). No evaluation of the data vector at all.

**What forces the saved derivatives to be recomputed.** Only what changes the knobs' responses: the fiducial point, the likelihood settings (accuracy, options, parameter files) and the code. The fingerprint of Section 9 catches changes of the input; a change of the code needs a flag set by hand (Section 11). The recipe $\mathsf{M}$, the priors, the scale cut and the covariance file never force a recompute.

# 9 Derivative accuracy

**Why a derivative needs more than a $\chi^2$.** Every evaluation of $m$ carries numerical noise: the integration grids, the non-Limber distance grid and CAMB's tolerances change the vector by small amounts that do not vary smoothly with the parameters. A finite difference divides that noise by the step, so a setting that converges a $\chi^2$ (the MCMC criterion is $|\Delta\chi^2| < 0.2$) can leave a derivative noisy.

The noise biases the forecast one way. A column $d_j + e_j$, with $e_j$ its noise, gives $F_{jj} = (d_j + e_j)^\mathsf{T}\mathsf{C}^{-1}(d_j + e_j)$, whose average exceeds the true value by $\langle e_j^\mathsf{T}\mathsf{C}^{-1}e_j\rangle > 0$: noise looks like information, and the FoM comes out too high.

**The study: a starting configuration, and one control at a time.** The convergence study, `tests/studies/run_fisher_derivatives.py`, measures the effect on $\Delta z = 0.05$. It scores the marginalized errors of $w_0$, $w_a$, $\Omega_m$ and $A_s$, both FoMs, and the shifts under truths that carry a term the fit lacks or holds with a prior (the models are defined in the higher-order note, `notes/higher_order_bias`), at $k_{\max} = 0.075$ and $0.2\,h$/Mpc. The FoM offsets of the grid scan, the step test, the final settings and DerivKit are the paper model's, scored from the study's derivatives on the paper-fiducial cuts (the `data.json` behind Figure 4). The error and shift stabilities, and the offsets of the other controls (CAMB's boost among them), are scored on the study's own system, model B + tilts with $\alpha_2$ free and the exploration $\sigma_v$ widths; the text marks each such number. The derivative columns do not depend on the model, so both scorings test the same columns (Appendix B).

The scan starts from one configuration, the starting configuration: a grid factor $g = 128$ with every other control raised, CosmoLike's `accuracyboost` 4 and `integration_accuracy` 2, the low-rank engine's rank 8 (the number of terms kept in its sum) and multipole cap 2000, and CAMB's `AccuracyBoost` 2.0, `k_per_logint` 30 and maximum wavenumber $80\,\mathrm{Mpc}^{-1}$. The scan then varied one control at a time away from it. Here $g$ is the product `accuracyboost` $\times$ `nonlimber_accuracyboost`, and the non-Limber projection integrates over about $511\,g$ comoving distances; the MCMC value on $\Delta z = 0.05$ is $g = 32$. Each control was lowered toward its MCMC value; the grid was also raised to 256 and 512. The five-point stencil at step $h$ on $g = 512$ is the converged value every result is compared with. The final tier, chosen at the end of the section, uses $g = 256$. (The study's result files name the two endpoints by key: `reference` is the starting configuration and `candidate` the final settings.)

**The noise of one evaluation.** The computed vector is not the smooth model alone. Write

$$m_\mathrm{c}(\theta) \;=\; m(\theta) \;+\; n(\theta), \tag{27}$$

with $m$ the model, smooth in the parameters, and $n$ the numerical error of one evaluation, from the causes named in "Why a derivative needs more than a $\chi^2$": the integration grids, the non-Limber distance grid and CAMB's tolerances. $n$ is small, but it does not vary smoothly with $\theta$: a tiny move of the parameters shifts where the model falls on the fixed grids, and the error changes irregularly from point to point. The study measures the size of $n$ from the five stencil points themselves (`stencil_noise` in `tests/studies/run_fisher_derivatives.py`), with no extra evaluations. The next paragraphs build the estimator one step at a time.

**The fourth difference removes the smooth part.** Take the five equally spaced points $\theta_0 + k\epsilon$, $k = -2, \ldots, 2$, and combine them with the weights $c_k = (1, -4, 6, -4, 1)$, the binomial coefficients of $(1 - x)^4$ with alternating signs:

$$\Delta^4 m \;=\; \sum_{k=-2}^{2} c_k\, m(k) \;=\; m(-2) - 4\, m(-1) + 6\, m(0) - 4\, m(1) + m(2), \tag{28}$$

writing $m(k)$ for $m_{\mathrm{c}}(\theta_0 + k\epsilon)$. Expand each point of the smooth part in a Taylor series around $\theta_0$,

$$m(\theta_0 + k\epsilon) \;=\; \sum_{n=0}^{\infty} \frac{(k\epsilon)^n}{n!}\, m^{(n)}(\theta_0), \tag{29}$$

and collect the terms order by order: the $n$-th derivative enters $\Delta^4 m$ with the coefficient $(\epsilon^n/n!) \sum_k c_k\, k^n$. These sums can be checked by hand:

| $n$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| $\sum_k c_k\, k^n$ | 0 | 0 | 0 | 0 | 24 | 0 | 120 |

For $n = 4$ the sum is $16 - 4 + 0 - 4 + 16 = 24$; every odd order vanishes because the weights are symmetric, $c_{-k} = c_k$, while $k^n$ is odd. So everything up to the third derivative cancels exactly, the fourth order survives with $\epsilon^4\,(24/4!) = \epsilon^4$, and the next surviving order is $n = 6$ with $120/6! = 1/6$:

$$\Delta^4 m\big|_{\text{smooth}} \;=\; \epsilon^4\, m''''(\theta_0) \;+\; \frac{\epsilon^6}{6}\, m^{(6)}(\theta_0) \;+\; \ldots \tag{30}$$

The study's steps are a few percent of each parameter, and the model varies smoothly on the scale of the parameter itself, so $\epsilon^4 m''''$ is suppressed by $\epsilon^4$, roughly seven orders of magnitude: for a smooth vector the fourth difference is essentially zero. What survives in Eq. (28) is the noise.

**Independent noise adds in quadrature: the 70.** The noise part of the fourth difference is $\sum_k c_k\, n_k$, with $n_k$ the error at point $k$. Errors that change irregularly from point to point are modelled as independent draws with a common variance $\sigma^2$. The variance of a weighted sum of independent terms is the sum of the squared weights times $\sigma^2$:

$$\mathrm{Var}\Big(\sum_k c_k\, n_k\Big) \;=\; \sigma^2 \sum_k c_k^2 \;=\; (1 + 16 + 36 + 16 + 1)\,\sigma^2 \;=\; 70\,\sigma^2. \tag{31}$$

So $|\Delta^4 m|/\sqrt{70}$ estimates $\sigma$, the noise of one evaluation. One stencil is enough: the data vector has thousands of entries, the fourth difference is taken entry by entry, and the length of the result averages over all of them, so a single realization gives a tight estimate.

**A toy, with invented numbers.** Take one entry, the smooth model $m(\theta) = e^\theta$ at $\theta_0 = 0$ with $\epsilon = 0.01$, and plant noise of true rms $0.8 \times 10^{-6}$ at the five points: $n_k = (1.3, -0.4, 0.9, -0.6, 0.2) \times 10^{-6}$. The smooth part of the fourth difference is $\epsilon^4\, e^{\theta_0} = 1.0 \times 10^{-8}$, a thousand times below the noise combination. The noise part is $1.3 + (-4)(-0.4) + 6\,(0.9) + (-4)(-0.6) + 0.2 = 10.9$, times $10^{-6}$, so the estimate is $10.9 \times 10^{-6}/\sqrt{70} = 1.3 \times 10^{-6}$: the right size, though off the true $0.8 \times 10^{-6}$ by a factor 1.7, as a single entry must be, since one number is estimating a variance. Averaged over the thousands of entries of a real data vector, that scatter disappears.

**Measuring a vector in the data's own units.** A raw $|\Delta^4 m|$ would mix entries with very different error bars, and neighbouring band powers are correlated. The cure is to transform the vector first, so that, measured against the data's covariance $\mathsf{C}$, the error of every entry is one and no two entries are correlated. After the transformation each entry counts in units of its own error bar, and the plain sum of squares of the transformed vector is a $\chi^2$.

**The transformation, through the eigenvectors.** Diagonalize the covariance, $\mathsf{C} = U\Lambda U^{\mathsf{T}}$, with $U$ the orthogonal matrix of its eigenvectors and $\Lambda$ the diagonal matrix of the variances along them. For a vector $v$, the rotation $U^{\mathsf{T}}v$ gives its components along the eigenvectors, the directions in which the data's errors are uncorrelated, and $\Lambda^{-1/2}$ then divides each component by its standard deviation. So

$$y \;=\; \Lambda^{-1/2}\, U^{\mathsf{T}}\, v \tag{32}$$

has unit, uncorrelated errors, and $|y|^2 = v^{\mathsf{T}}U\Lambda^{-1}U^{\mathsf{T}}v = v^{\mathsf{T}}\mathsf{C}^{-1}v$. This transformation is called *whitening*, after white noise: noise with equal power in every direction, as white light holds every colour.

**Why a Cholesky factor does the same job.** The code whitens with the Cholesky factorization instead, $\mathsf{C} = LL^{\mathsf{T}}$ with $L$ lower triangular (a matrix analogue of the square root): $y = L^{-1}v$ has covariance $L^{-1}\,\mathsf{C}\,L^{-\mathsf{T}} = 1$, so its errors are also unit and uncorrelated, and its length is the same, $|L^{-1}v|^2 = v^{\mathsf{T}}\mathsf{C}^{-1}v$. The two transforms differ only by a rotation, which changes no length; the Cholesky factorization is cheaper and numerically stable, so the code uses it.

**A $2\times 2$ toy.** Two band powers with error bars 1 and 2 and correlation 0.5 have

$$\mathsf{C} = \begin{pmatrix} 1 & 1 \\ 1 & 4 \end{pmatrix}, \qquad L = \begin{pmatrix} 1 & 0 \\ 1 & \sqrt{3} \end{pmatrix}, \qquad L^{-1} = \begin{pmatrix} 1 & 0 \\ -1/\sqrt{3} & 1/\sqrt{3} \end{pmatrix} \tag{33}$$

(the off-diagonal entry is $0.5\times 1\times 2 = 1$; multiplying out $LL^{\mathsf{T}}$ recovers $\mathsf{C}$). Take the difference vector $v = (1,1)$: whitened, $y = L^{-1}v = (1,0)$, and $|y|^2 = 1$. Directly, $\mathsf{C}^{-1} = \frac{1}{3}\left(\begin{smallmatrix} 4 & -1 \\ -1 & 1 \end{smallmatrix}\right)$ and $v^{\mathsf{T}}\mathsf{C}^{-1}v = (4-1-1+1)/3 = 1$: the same $\chi^2$. Counting each entry against its own error bar alone would give $1/1 + 1/4 = 1.25$; the two entries are correlated, so part of the misfit repeats and whitening counts it once. The eigenvector route, with variances $(5 \pm \sqrt{13})/2 \approx 4.30$ and 0.70, gives a different $y$ of the same length 1: the two transforms differ by a rotation.

**The estimate in these units.** The study whitens the fourth difference with the shipped-mask covariance at the survey area ($f_A$ applied) and reports

$$\hat{\sigma} \;=\; \left|L^{-1}\,\Delta^4 m\right| / \sqrt{70}\,: \tag{34}$$

the numerical error of one evaluation, measured in the data's own error bars ($\sqrt{\chi^2}$ units), listed per stencil parameter in Table 9. Reading one entry: $1.4\times 10^{-2}$ for $w_a$ at the MCMC settings means that the numerical error of one evaluation along the $w_a$ stencil moves the whitened vector by 0.014, a $\Delta\chi^2$ of $2\times 10^{-4}$ – far below the MCMC convergence criterion $|\Delta\chi^2| < 0.2$, and harmless for a likelihood evaluation. The next paragraph is why it still matters.

| configuration | $10^9 A_s$ | $H_0$ | $\Omega_m$ | $w_0$ | $w_a$ |
|---|---|---|---|---|---|
| MCMC settings | $6.0 \times 10^{-6}$ | $2.1 \times 10^{-3}$ | $1.7 \times 10^{-2}$ | $2.5 \times 10^{-2}$ | $1.4 \times 10^{-2}$ |
| $g = 64$ | $4.5 \times 10^{-6}$ | $2.2 \times 10^{-3}$ | $8.1 \times 10^{-3}$ | $1.1 \times 10^{-2}$ | $4.8 \times 10^{-3}$ |
| $g = 128$ (starting) | $4.5 \times 10^{-6}$ | $2.2 \times 10^{-3}$ | $6.5 \times 10^{-3}$ | $4.3 \times 10^{-3}$ | $1.8 \times 10^{-3}$ |
| $g = 256$ | $4.5 \times 10^{-6}$ | $2.2 \times 10^{-3}$ | $6.1 \times 10^{-3}$ | $2.2 \times 10^{-3}$ | $3.5 \times 10^{-4}$ |
| final settings | $4.6 \times 10^{-6}$ | $3.4 \times 10^{-3}$ | $3.1 \times 10^{-3}$ | $5.8 \times 10^{-4}$ | $3.2 \times 10^{-4}$ |

Table 9: The noise of one evaluation along each stencil parameter, in $\sqrt{\chi^2}$ units, on $\Delta z = 0.05$. The rows with a grid factor $g$ keep every other control at the starting configuration's value ($g = 128$ is the starting configuration itself); the final settings change the other controls as well, at the final grid factor $g = 256$. The five columns shown are the stencil parameters the FoM pairs read most; the other nine ($n_s$, $\Omega_b$, the seven $\sigma_v$) are measured alongside and enter the prediction of Table 13. The unit is the natural one for a Fisher forecast: the data vector enters $F = \mathsf{D}^\mathsf{T}\mathsf{C}^{-1}\mathsf{D}$ only through the metric $\mathsf{C}^{-1}$, so a change $\delta m$ matters only through its whitened length $\sqrt{\delta m^\mathsf{T}\mathsf{C}^{-1}\delta m}$, and a column whose noise is $\sigma$ per evaluation adds about $0.9\,(\sigma/\epsilon)^2$ to $F_{jj}$, Eq. (37).

**Why the jitter matters for a derivative.** First the objects. The Fisher matrix, Eq. (2), is built from the derivative matrix $\mathsf{D}$, one column per sampled parameter $p_j$. The column the code wants is $d_j = \partial m/\partial p_j$ at the fiducial point: the exact derivative of the smooth model. The column it gets applies the stencil, Eq. (14), to the computed vectors of Eq. (27), $m_\mathrm{c} = m + n$. The smooth parts combine to $d_j$ (up to the $\epsilon^4$ truncation of the stencil), the noise parts to an error, so the computed column is $d_j + e_j$ with

$$e_j \;=\; \frac{n(-2) - 8\,n(-1) + 8\,n(1) - n(2)}{12\,\epsilon_j}, \tag{35}$$

writing $n(k)$ for the evaluation noise $k$ steps from the fiducial along $p_j$, and $\epsilon_j$ for that parameter's step. Like $d_j$, the error $e_j$ is a vector with one entry per band power. And each column has its own: the four evaluations behind the column of another parameter $p_k$ move only $p_k$ (the center point carries weight zero and is never evaluated), so $e_k$ is built from its own evaluations, independent of those behind $e_j$.

What the noise does to $F$. Put the computed column into Eq. (2):

$$F_{jj} \;=\; (d_j + e_j)^\mathsf{T}\mathsf{C}^{-1}(d_j + e_j) \;=\; d_j^\mathsf{T}\mathsf{C}^{-1}d_j \;+\; 2\,d_j^\mathsf{T}\mathsf{C}^{-1}e_j \;+\; e_j^\mathsf{T}\mathsf{C}^{-1}e_j. \tag{36}$$

The first term is the true entry. The middle term is linear in the noise, and the noise scatters with no preferred sign (the independent draws of "Independent noise adds in quadrature: the 70"), so it averages to zero. The last term is a sum of squares, like a $\chi^2$: positive for every noise realization, so on average the noise only *adds* to $F_{jj}$ – the one-way bias of "Why a derivative needs more than a $\chi^2$": noise looks like information. An off-diagonal entry $F_{jk}$ gains $e_j^\mathsf{T}\mathsf{C}^{-1}e_k$ and two linear cross terms; $e_j$ and $e_k$ come from separate evaluations, so all three average to zero. The systematic push sits on the diagonal.

How large the push is. The four errors in the numerator of Eq. (35) are independent, so their whitened lengths add in quadrature through the squared weights, $1 + 64 + 64 + 1 = 130$, and the whitened length of $e_j$ is about

$$\frac{\sqrt{1 + 64 + 64 + 1}}{12}\,\frac{\sigma}{\epsilon_j} \;=\; \frac{\sqrt{130}}{12}\,\frac{\sigma}{\epsilon_j} \;\approx\; 0.95\,\frac{\sigma}{\epsilon_j}, \tag{37}$$

| | $w_a$ | | $10^9 A_s$ | |
|---|---|---|---|---|
| | $k_{\max} = 0.075$ | 0.2 | 0.075 | 0.2 |
| $F_{jj}$ (others fixed) | 670 | 2000 | $2.5 \times 10^4$ | $5.9 \times 10^4$ |
| $1/\sigma_j^2$ (marginalized) | 0.013 | 0.18 | 0.048 | 0.87 |
| $a_j$, MCMC settings | 0.41 | | $1.8 \times 10^{-8}$ | |
| $a_j$, final settings | $2.4 \times 10^{-4}$ | | $1.1 \times 10^{-8}$ | |
| $a_j\sigma_j^2$, MCMC settings | 33 | 2.3 | $3.8 \times 10^{-7}$ | $2.1 \times 10^{-8}$ |
| $a_j\sigma_j^2$, final settings | 0.019 | $1.3 \times 10^{-3}$ | $2.3 \times 10^{-7}$ | $1.3 \times 10^{-8}$ |

Table 10: The spurious additions against the real information on $w_a$ and $10^9 A_s$, on the converged $\Delta z = 0.05$ paper-model system. $F_{jj}$: the information with every other parameter fixed. $1/\sigma_j^2$: the marginalized information. $a_j = 0.9\,(\sigma/\epsilon_j)^2$: the spurious addition, computed from the unrounded noise behind Table 9; it does not depend on the cut. $a_j\sigma_j^2$: the addition over the marginalized information, the dimensionless size. $k_{\max}$ in $h$/Mpc, quoted at $z = 0$.

with $\sigma$ the whitened ($\sqrt{\chi^2}$-unit) noise of one evaluation along that column's stencil, Table 9: a derivative divides the jitter by its step. Squaring gives the average of the spurious term of Eq. (36), $\langle e_j^{\mathsf{T}} \mathsf{C}^{-1} e_j \rangle \approx 0.9\,(\sigma/\epsilon_j)^2$; with $\sigma$ in $\sqrt{\chi^2}$ units and $\epsilon_j$ in the parameter's units, the addition is in one over the parameter's units squared, the units of $F_{jj}$.

In numbers, along $w_a$. Its fiducial value is zero and its prior flat, so its step is absolute and unscaled, $\epsilon_{w_a} = h = 0.02$ ("The cosmological and $\sigma_v$ columns"). At the MCMC settings the $w_a$ noise is $\sigma = 1.4 \times 10^{-2}$, Table 9, so the column adds a spurious $0.9\,(0.014/0.02)^2 \approx 0.44$ to $F_{w_a w_a}$ ($w_a$ is dimensionless, so the entry is a plain number). At the final grid, $\sigma = 3.5 \times 10^{-4}$, the addition is $0.9\,(3.5 \times 10^{-4}/0.02)^2 \approx 2.8 \times 10^{-4}$. The spurious information grows as the noise *squared*: the fortyfold fall of the $w_a$ noise in Table 9, from the MCMC settings to the final grid, is a 1600-fold fall of its spurious contribution to $F$.

And along $10^9 A_s$, the quietest column. Its noise is $4.6 \times 10^{-6}$ at the final settings on a step $\epsilon = 0.02 \times 2.1 = 0.042$, so the addition is $0.9\,(4.6 \times 10^{-6}/0.042)^2 \approx 1 \times 10^{-8}$ – negligible. FoM$(\Omega_m, 10^9 A_s)$ therefore picks up evaluation noise through the $\Omega_m$ column and the other columns correlated with the pair, not through $A_s$ itself.

Big or small? An addition needs a base, and Table 10 puts the two columns' additions next to the information the data actually carry, on the converged $\Delta z = 0.05$ paper-model system. Two bases offer themselves and they differ by factors of $10^4$ to $10^5$: the diagonal entry $F_{jj}$, the information on $p_j$ with every other parameter held fixed, and the marginalized information $1/\sigma_j^2$, with $\sigma_j$ the forecast error when everything else is free. The right yardstick is the marginalized one: the degeneracies make $1/\sigma_j^2$ far smaller than $F_{jj}$, and the fake prior competes with what is left after marginalization, not with the fixed-parameter information. By that yardstick the MCMC-settings addition is about 30 times the real marginalized information on $w_a$ at $k_{\max} = 0.075\,h$/Mpc – the size the MCMC row of the measured-against-predicted comparison below registers – while the final-settings addition is about 2% of it, and the $10^9 A_s$ additions never exceed a few times $10^{-7}$ of theirs. The subsection below turns exactly this ratio, $a_j\sigma_j^2 = (\sigma_j/w_j)^2$, into the measure of the fake prior. A rule of thumb for what follows: once $a_j\sigma_j^2$ is below about $10^{-2}$ for every column, the noise moves the FoM by less than half a percent – that setting has settled.

**Choosing the settings cheaply: how the evaluation noise reaches the FoM.** The problem this subsection solves. Finding the accuracy settings at which the FoM stops changing normally takes a full derivative base at every candidate setting: 154 evaluations (Table 6) for the twenty sampled parameters plus the per-bin columns – about 50 minutes on $\Delta z = 0.05$ at the final settings (some 20 s per evaluation) and 1.8 to 2.1 hours per base on the thin layouts, so a scan over candidate settings costs hours to days. This subsection builds a cheap estimate instead: one low-accuracy base, computed once, and then, per candidate setting, only the noise of a few heavy columns – about 8 to 20 evaluations.

First the question. The paragraphs above pin down what the noise does to the Fisher matrix: the noise of stencil column $j$ adds, on average, about $0.9\,(\sigma/\epsilon_j)^2$ to the diagonal entry $F_{jj}$, and nothing, on average, anywhere else. But the published numbers are not Fisher entries; they are marginalized errors and FoMs. What does the addition do to $\mathrm{FoM}(w_0, w_a)$?

What the FoM reads. Recall Section 3: $F^{-1}$ is the forecast covariance of the parameters, $\mathsf{C}_w$ is its $2 \times 2$ block for the pair $(w_0, w_a)$, and $\mathrm{FoM}(w_0, w_a) = [\det \mathsf{C}_w]^{-1/2}$ is the inverse area of the pair's error ellipse when every other parameter is free. So the additions land in $F$, while the FoM reads a block of $F^{-1}$.

Why one cannot just look at $F_{ww}$. A block of the inverse is not the inverse of the block: $(F^{-1})_{ww} \neq (F_{ww})^{-1}$ whenever the pair correlates with any other parameter, because inverting $F$ mixes every parameter into every block – that mixing *is* marginalization. The additions must therefore be followed through the inverse of the full matrix. The route below, each step a piece of the tool: the addition rewritten as a prior ($a_j$, the one ingredient a setting changes); the marginalization as a block formula (where the setting-independent $\sigma_j$ enter); a $3 \times 3$ determinant and a worked toy (the estimate in its two-column form); and the general case of every column noisy at once (the weights $R_j^2$, and with them the low cost).

The noise as a fake prior. The addition has a familiar form. Below Eq. (2): a Gaussian prior of width $w$ adds $1/w^2$ to its parameter's diagonal entry of $F$, and nothing else. The noise of column $j$ adds, on average,

$$a_j \;=\; \langle e_j^{\mathsf{T}} \mathsf{C}^{-1} e_j \rangle \;\approx\; 0.9\,(\sigma/\epsilon_j)^2 \tag{38}$$

to $F_{jj}$, and nothing, on average, anywhere else (the cross terms of "What the noise does to $F$" have zero mean). Matching the two statements: the noise of column $j$ acts exactly like a Gaussian prior on $p_j$ with $1/w_j^2 = a_j$ – a spurious prior of width $w_j = a_j^{-1/2} \approx \epsilon_j/(0.95\,\sigma)$, information the data do not carry. For the tool, $a_j$ is the one ingredient that depends on the candidate setting, and it is cheap: four evaluations along column $j$ measure it.

In numbers, along $w_a$. At the MCMC settings the addition computed above is $a_{w_a} \approx 0.44$, a spurious prior of width $1/\sqrt{0.44} \approx 1.5$ on $w_a$; at the final grid, $a_{w_a} \approx 2.8 \times 10^{-4}$, a width of 60. Whether a prior of width 1.5 matters depends on how well the data measure $w_a$: a forecast feels a prior through the ratio of the parameter's marginalized error $\sigma_j$ to the prior's width. The determinant expansion below makes that precise.

**Marginalizing over the other parameters.** Why the tool needs this step: the noise reaches the FoM only through the marginalized block, so the estimate will be built from marginalized quantities – $\sigma_j$ here, $R_j$ later – which barely depend on the accuracy setting and can be computed once, on a cheap system.

The block split. Order the sampled parameters so the pair $w = (w_0, w_a)$ comes first and the rest – call them $o$, for "others" – follow, and write the Fisher matrix in blocks,

$$F \;=\; \begin{pmatrix} F_{ww} & F_{wo} \\ F_{ow} & F_{oo} \end{pmatrix} :$$

$F_{ww}$ is the pair's $2 \times 2$ corner, $F_{oo}$ the square block of the other parameters among themselves, and $F_{wo} = F_{ow}^{\mathsf{T}}$ (two rows, one column per other parameter) their cross information. The pair's block of $F^{-1}$, the $\mathsf{C}_w$ of the FoM, is the inverse of

$$F_w \;=\; F_{ww} - F_{wo}\, F_{oo}^{-1}\, F_{ow}, \qquad \mathsf{C}_w = F_w^{-1} : \tag{39}$$

$F_w$ is the $2 \times 2$ information left on the pair when every other parameter is free – the subtraction is what marginalization costs – and $\mathrm{FoM}(w_0, w_a) = [\det \mathsf{C}_w]^{-1/2} = (\det F_w)^{1/2}$.

Where the formula comes from. Eq. (39) is a standard linear-algebra result, the inverse of a partitioned matrix: split any invertible matrix into four blocks,

$$M \;=\; \begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{pmatrix} \quad\Longrightarrow\quad (M^{-1})_{11} \;=\; \left(M_{11} - M_{12}\, M_{22}^{-1}\, M_{21}\right)^{-1},$$

whenever $M_{22}$ is invertible too. The combination $M_{11} - M_{12}\, M_{22}^{-1}\, M_{21}$ is called the Schur complement of $M_{22}$ in $M$; block matrices and their inverses are collected in the Matrix Cookbook [4]. Here $M_{11} = F_{ww}$, $M_{12} = F_{wo}$, $M_{21} = F_{ow}$ ($F$ is symmetric), $M_{22} = F_{oo}$, and $\mathsf{C}_w^{-1} = F_w$ is the Schur complement of $F_{oo}$.

The proof, in three steps on $F$ itself. First, name the unknowns: the first block column of $F^{-1}$ holds $\mathsf{C}_w$ on top and, below it, the cross block $(F^{-1})_{ow}$, the forecast covariance of the other parameters with the pair. Second, read $F\, F^{-1} = I$ block row by block row on that column ($I$ the identity):

$$F_{ww}\, \mathsf{C}_w + F_{wo}\, (F^{-1})_{ow} \;=\; I, \qquad F_{ow}\, \mathsf{C}_w + F_{oo}\, (F^{-1})_{ow} \;=\; 0.$$

Third, solve: the second equation gives $(F^{-1})_{ow} = -F_{oo}^{-1}\, F_{ow}\, \mathsf{C}_w$, and substituting it into the first leaves $\left(F_{ww} - F_{wo}\, F_{oo}^{-1}\, F_{ow}\right) \mathsf{C}_w = I$, which is Eq. (39).

The statistics behind the algebra. $F_{ww}$ alone is the information on $(w_0, w_a)$ with every other parameter held fixed, and the subtracted $F_{wo}\, F_{oo}^{-1}\, F_{ow}$ is the information the other parameters absorb when they are left free, that is, marginalized. Marginalizing a Gaussian keeps a block of its covariance $F^{-1}$ (Section 3); written in terms of $F$, that block is this Schur complement.

**Why the additions pass through the marginalization whole.** The claim, then its construction. Let only the $w_0$ and $w_a$ columns be noisy. Their additions land on the diagonal of $F_w$ unchanged – exactly, not only to first order. This is what keeps the estimate to one line: no new matrix algebra per setting. Three block steps show it, and a $3 \times 3$ example then shows it entry by entry.

Step one, the noisy matrix in blocks. On average the noise adds $a_{w_0}$ and $a_{w_a}$ to the two diagonal entries of the pair and nothing else (the off-diagonal additions average to zero, as shown above), so the noisy Fisher matrix is

$$F' \;=\; F + \mathsf{E}, \qquad \mathsf{E} \;=\; \begin{pmatrix} \mathsf{A} & 0 \\ 0 & 0 \end{pmatrix}, \qquad \mathsf{A} \;=\; \begin{pmatrix} a_{w_0} & 0 \\ 0 & a_{w_a} \end{pmatrix},$$

with zeros in every block the noise does not touch.

Step two, the blocks of $F'$. Adding $\mathsf{E}$ changes one block and leaves three alone:

$$F'_{ww} = F_{ww} + \mathsf{A}, \qquad F'_{wo} = F_{wo}, \qquad F'_{ow} = F_{ow}, \qquad F'_{oo} = F_{oo}.$$

Step three, into the marginalization. Put these blocks into Eq. (39):

$$F'_w \;=\; (F_{ww} + \mathsf{A}) - F_{wo}\, F_{oo}^{-1}\, F_{ow} \;=\; F_w + \mathsf{A}.$$

No approximation was made: $F_{ww}$ is the one block that enters Eq. (39) linearly and whole, and the subtracted term is built from $F_{wo}$ and $F_{oo}$ alone, which the noise leaves untouched.

The same, entry by entry. Take three parameters $(w_0, w_a, o)$, so $F_{oo} = f_{33}$ is a number, and write the marginalization out:

$$F \;=\; \begin{pmatrix} f_{11} & f_{12} & f_{13} \\ f_{12} & f_{22} & f_{23} \\ f_{13} & f_{23} & f_{33} \end{pmatrix} \quad\Longrightarrow\quad F_w \;=\; \begin{pmatrix} f_{11} - f_{13}^2/f_{33} & f_{12} - f_{13}f_{23}/f_{33} \\ f_{12} - f_{13}f_{23}/f_{33} & f_{22} - f_{23}^2/f_{33} \end{pmatrix}.$$

The noise changes only $f_{11} \to f_{11} + a_{w_0}$ and $f_{22} \to f_{22} + a_{w_a}$:

$$F' \;=\; \begin{pmatrix} f_{11} + a_{w_0} & f_{12} & f_{13} \\ f_{12} & f_{22} + a_{w_a} & f_{23} \\ f_{13} & f_{23} & f_{33} \end{pmatrix}$$

$$\Longrightarrow\quad F'_w \;=\; \begin{pmatrix} f_{11} + a_{w_0} - f_{13}^2/f_{33} & f_{12} - f_{13}f_{23}/f_{33} \\ f_{12} - f_{13}f_{23}/f_{33} & f_{22} + a_{w_a} - f_{23}^2/f_{33} \end{pmatrix}.$$

None of the three subtracted pieces – $f_{13}^2/f_{33}$, $f_{13}f_{23}/f_{33}$, $f_{23}^2/f_{33}$ – contains $f_{11}$ or $f_{22}$, so $F'_w = F_w + \mathsf{A}$: each addition rides through the marginalization into its own diagonal entry.

**A worked toy, part one: the marginalization.** The toy the rest of this subsection reuses. Three sampled parameters, the pair $(w_0, w_a)$ and one other parameter $o$ standing in for all the others, with the round-number Fisher matrix

$$F \;=\; \begin{pmatrix} 5 & 4 & 1 \\ 4 & 5 & 2 \\ 1 & 2 & 3 \end{pmatrix}$$

in that order (all three parameters treated as dimensionless, so every entry is a plain number; the matrix is positive definite, with eigenvalues 0.71, 2.6 and 9.7). The entries are chosen so the forecast resembles the real one: a strong $w_0$–$w_a$ anticorrelation and a sizeable correlation with $o$, computed below.

The marginalization, with the entry-by-entry formula. Here $f_{33} = 3$, and the three subtracted pieces are $f_{13}^2/f_{33} = 1/3$, $f_{13}f_{23}/f_{33} = 2/3$ and $f_{23}^2/f_{33} = 4/3$:

$$F_w \;=\; \begin{pmatrix} 5 - 1/3 & 4 - 2/3 \\ 4 - 2/3 & 5 - 4/3 \end{pmatrix} \;=\; \begin{pmatrix} 14/3 & 10/3 \\ 10/3 & 11/3 \end{pmatrix} \;\approx\; \begin{pmatrix} 4.67 & 3.33 \\ 3.33 & 3.67 \end{pmatrix},$$

with $\det F_w = (14 \times 11 - 10 \times 10)/9 = 6$, so FoM $= \sqrt{6} \approx 2.45$.

Fixed versus free. The same matrices show what the marginalization costs. With $o$ held fixed, the pair's information is the corner block $F_{ww}$ itself: $\det F_{ww} = 5 \times 5 - 4 \times 4 = 9$, the FoM would be $\sqrt{9} = 3.00$, and the errors $\sigma_{w_0} = \sigma_{w_a} = \sqrt{5/9} \approx 0.745$. With $o$ free, $\det F_w = 6$: FoM $= \sqrt{6} \approx 2.45$, and the marginalized errors grow to $\sigma_{w_0} \approx 0.782$ and $\sigma_{w_a} \approx 0.882$ (part two below inverts $F_w$ in full). The difference is the subtracted matrix,

$$F_{wo}\, F_{oo}^{-1}\, F_{ow} \;=\; \frac{1}{3}\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix},$$

the information $o$ absorbs when it is left free – and it is why the FoM cannot be read off $F_{ww}$.

The pass-through, with numbers. Add $a_{w_0} = 0.5$ and $a_{w_a} = 0.3$. The direct route – form $F + \mathsf{E}$, redo the Schur complement with $f_{11} = 5.5$ and $f_{22} = 5.3$ – gives

$$F'_w \;=\; \begin{pmatrix} 5.5 - 1/3 & 4 - 2/3 \\ 4 - 2/3 & 5.3 - 4/3 \end{pmatrix} \;\approx\; \begin{pmatrix} 5.17 & 3.33 \\ 3.33 & 3.97 \end{pmatrix} \;=\; F_w + \begin{pmatrix} 0.5 & 0 \\ 0 & 0.3 \end{pmatrix},$$

the additions sitting whole on the diagonal, as the blocks promised. How much the FoM moved takes the determinant, next.

**The two-column result, step by step.** This paragraph is the estimate in its two-column form: a setting-independent weight, $\sigma_j^2$, times the setting-dependent $a_j$.

The FoM as one determinant. Stay with the $3 \times 3$ matrix of the construction above: the pair's rows first, one other parameter $o$, entries $f_{11} \ldots f_{33}$. The block result has a determinant form too, $\det F = \det F_{oo}\, \det F_w$ [4]; with a single other parameter $\det F_{oo} = f_{33}$, so $\mathrm{FoM}^2 = \det F_w = \det F / f_{33}$ (in the toy, $\det F_w = 18/3 = 6$). The noise leaves $f_{33}$ untouched, so that factor cancels in the ratio:

$$\frac{\mathrm{FoM}'^2}{\mathrm{FoM}^2} \;=\; \frac{\det F'_w}{\det F_w} \;=\; \frac{\det F'}{\det F}.$$

The determinant, term by term. A determinant is linear in each row separately, and the noise touches one entry of row 1 and one of row 2:

$$\begin{aligned} \det F' \;&=\; \det \begin{pmatrix} f_{11} + a_{w_0} & f_{12} & f_{13} \\ f_{12} & f_{22} + a_{w_a} & f_{23} \\ f_{13} & f_{23} & f_{33} \end{pmatrix} \\ &=\; \det F \;+\; a_{w_0}\,(f_{22} f_{33} - f_{23}^2) \;+\; a_{w_a}\,(f_{11} f_{33} - f_{13}^2) \;+\; a_{w_0}\, a_{w_a}\, f_{33}. \end{aligned}$$

Each coefficient is a minor of $F$: $f_{22} f_{33} - f_{23}^2$ is what is left after deleting row and column 1, $f_{11} f_{33} - f_{13}^2$ after deleting row and column 2, and $f_{33}$ after deleting both.

Each term in forecast language. The cofactor formula for an inverse – the adjugate rule behind Cramer's rule, $(F^{-1})_{ii} = (\text{the minor of the } (i,i) \text{ entry}) / \det F$ – names all three coefficients once they are divided by $\det F$:

$$\frac{f_{22} f_{33} - f_{23}^2}{\det F} \;=\; (F^{-1})_{11} \;=\; \sigma_{w_0}^2, \qquad \frac{f_{11} f_{33} - f_{13}^2}{\det F} \;=\; \sigma_{w_a}^2, \qquad \frac{f_{33}}{\det F} \;=\; \frac{1}{\det F_w} \;=\; \det \mathsf{C}_w,$$

the pair's marginalized variances and the determinant of its forecast covariance, for which $\det \mathsf{C}_w = \sigma_{w_0}^2 \sigma_{w_a}^2 - (\mathsf{C}_w)_{12}^2 = \sigma_{w_0}^2 \sigma_{w_a}^2\,(1 - \rho^2)$, with $\rho = (\mathsf{C}_w)_{12} / (\sigma_{w_0} \sigma_{w_a})$ the forecast correlation of $w_0$ and $w_a$. In the toy the three numbers are $11/18 = 0.611$, $14/18 = 0.778$ and $3/18 = 1/6$ – the values part two below confirms from the explicit inverse.

The result. Divide the expansion by $\det F$, use the three names, and take the square root:

$$\frac{\mathrm{FoM}'}{\mathrm{FoM}} \;=\; \left[\frac{\det F'_w}{\det F_w}\right]^{1/2} \;=\; \left[\,1 + a_{w_0}\sigma^2_{w_0} + a_{w_a}\sigma^2_{w_a} + (1-\rho^2)\,a_{w_0}\sigma^2_{w_0}\,a_{w_a}\sigma^2_{w_a}\,\right]^{1/2}, \qquad (40)$$

the prime marking the noisy forecast, and the relation is exact. Each term $a_j\sigma_j^2 = (\sigma_j/w_j)^2$ is the squared ratio of the parameter's marginalized error to its spurious prior width: the dimensionless measure of how informative the fake prior is. (Careful with the two sigmas: $\sigma_j$ here is the forecast error of parameter $j$, not the evaluation noise $\sigma$ of Table 9; the evaluation noise lives inside $a_j$.) In the toy, $\det F' = 18 + 0.5\times 11 + 0.3\times 14 + 0.5\times 0.3\times 3 = 28.15$, so $\det F'/\det F = 1.564$ and $\mathrm{FoM}'/\mathrm{FoM} = 1.251$ – the same numbers as the Schur route through $F'_w$ of part one: the two routes agree. And the general case costs nothing new: with $m$ other parameters the linear coefficients become the cofactors of the $(1,1)$ and $(2,2)$ entries, the cross term's $f_{33}$ becomes the complementary minor $\det F_{oo}$, and every step above goes through unchanged.

The first-order rule. When both terms are small, $\sqrt{1+x} \approx 1 + x/2$ and the product term is second order, so

$$\frac{\mathrm{FoM}'}{\mathrm{FoM}} \;\approx\; 1 + \tfrac{1}{2}\,a_{w_0}\sigma^2_{w_0} + \tfrac{1}{2}\,a_{w_a}\sigma^2_{w_a}\,:$$

one first-order term per noisy column, not one per evaluation – the four evaluations behind a column already sit inside its $a_j$, through the squared stencil weights $(1+64+64+1)/144 \approx 0.9$ of Eq. (37). With equal terms, $x = a_{w_0}\sigma^2_{w_0} = a_{w_a}\sigma^2_{w_a}$, Eq. (40) is $[1+2x+(1-\rho^2)\,x^2]^{1/2} \approx 1+x$, and exactly $1+x$ at $\rho = 0$.

First numbers, with the real $w_a$ column. Let only the $w_a$ column be noisy and, for illustration alone, take $\sigma_{w_a} = 1$. At the MCMC settings, $a_{w_a} \approx 0.44$: $\mathrm{FoM}'/\mathrm{FoM} = (1+0.44)^{1/2} = 1.20$, a 20% spurious gain. At the final grid, $a_{w_a} \approx 2.8\times 10^{-4}$: $1 + 1.4\times 10^{-4}$.

**The toy, part two: the forecast errors and the noisy FoM.** The forecast errors. The $2\times 2$ inverse of the toy's $F_w$ is

$$\mathsf{C}_w \;=\; F_w^{-1} \;=\; \frac{1}{6}\begin{pmatrix} 11/3 & -10/3 \\ -10/3 & 14/3 \end{pmatrix} \;\approx\; \begin{pmatrix} 0.611 & -0.556 \\ -0.556 & 0.778 \end{pmatrix},$$

so $\sigma_{w_0} = \sqrt{0.611} \approx 0.78$, $\sigma_{w_a} = \sqrt{0.778} \approx 0.88$, and $\rho = -0.556/(0.78\times 0.88) \approx -0.81$: the strong anticorrelation announced $(1-\rho^2 \approx 0.35)$. For later use, the full inverse is

$$F^{-1} \;=\; \frac{1}{18}\begin{pmatrix} 11 & -10 & 3 \\ -10 & 14 & -6 \\ 3 & -6 & 9 \end{pmatrix}$$

(its top-left $2\times 2$ block is $\mathsf{C}_w$, as Eq. (39) promises), so $\sigma_o^2 = 9/18 = 0.5$, and $o$'s forecast correlations are 0.30 with $w_0$ and $-0.53$ with $w_a$.

The toy's three cases, collected. Table 11 runs the toy through three noise cases. The first is part one's tens-of-percent additions, $a_{w_0} = 0.5$ and $a_{w_a} = 0.3$: the terms of Eq. (40) are $a_{w_0}\sigma^2_{w_0} = 0.306$, $a_{w_a}\sigma^2_{w_a} = 0.233$ and the product term 0.025, and their sum 1.564 is the same number as $\det F'_w/\det F_w$ from part one's matrices. Once the terms are not small the first-order rule overshoots, and only Eq. (40) itself is exact. The second case shrinks the additions

| noise added | exact FoM′/FoM | first order |
|---|---|---|
| $a_{w_0} = 0.5$, $a_{w_a} = 0.3$ | 1.251 | 1.269 |
| $a_{w_0} = a_{w_a} = 0.05$ | 1.034 | 1.035 |
| $a_o = 0.6$ | 1.041 | 1.050 |

Table 11: The toy's three noise cases: the exact FoM ratio against the first-order rule – Eq. (40) expanded to first order for the pair's columns, Eq. (41) for the $o$ column.

to $a_{w_0} = a_{w_a} = 0.05$: at the few percent the published forecasts care about, first order is already the whole story.

Noise on the other column, the third case. Add $a_o = 0.6$ to the $o$ column alone. This addition sits in $f_{33}$, inside every subtracted piece of the entry-by-entry formula, so it does not pass through whole; it acts through the marginalization: redoing the Schur complement with $f_{33} = 3.6$ gives $\det F'_w = 6.50$ and the table's ratio $(6.50/6)^{1/2} = 1.041$ – a 4.1% spurious gain from a column that is neither $w_0$ nor $w_a$. The mechanism: the fake prior makes $o$ look better measured, a better-measured $o$ absorbs less information, and the pair keeps more. The next paragraph derives the first-order rule that predicts this number from $o$'s correlation with the pair.

**Every noisy column, through its correlation with the pair.** This paragraph is the estimate in full, Eq. (41) below: one weight $R_j^2\sigma_j^2$ per column, computed once, times each setting's $a_j$.

The general case, to first order. In the real system every column is noisy at once: the $2 \times 2$ matrix $\mathsf{A}$ of the construction grows to the full diagonal, $\mathsf{A} = \mathrm{diag}(a_1, a_2, \dots)$ with one $a_j$ per sampled parameter, and the noisy Fisher matrix is $F + \mathsf{A}$. Near the final settings every $a_j\sigma_j^2$ is small, so first order in $\mathsf{A}$ is enough. Three steps: how the inverse moves, which block to keep, and how a determinant responds.

How the inverse moves. For a small $\mathsf{A}$, factor and expand:

$$(F+\mathsf{A})^{-1} \;=\; \big[F\,(I+F^{-1}\mathsf{A})\big]^{-1} \;=\; (I+F^{-1}\mathsf{A})^{-1}\,F^{-1} \;\approx\; (I-F^{-1}\mathsf{A})\,F^{-1} \;=\; F^{-1}-F^{-1}\mathsf{A}\,F^{-1},$$

using $(I + X)^{-1} \approx I - X$ for small $X$, the matrix version of $1/(1 + x) \approx 1 - x$. So the change of the inverse is minus the change, sandwiched between two inverses, $\delta(F^{-1}) = -F^{-1}\,\mathsf{A}\,F^{-1}$ – the matrix version of $\delta(1/x) = -\delta x/x^2$.

Which block to keep. The FoM reads the pair's block of $F^{-1}$. Two pieces of notation first, used to the end of the derivation. Greek indices $\alpha, \beta$ run over the pair's two parameters $(w_0, w_a)$ only; the Latin index $j$ runs over all sampled parameters. And $(F^{-1})_{w,j}$ is parameter $j$'s column of $F^{-1}$ restricted to the pair's rows – a column of two numbers, entries $(F^{-1})_{\alpha j}$ – while $(F^{-1})_{j,w}$ is $j$'s row of $F^{-1}$ restricted to the pair's columns, the same two numbers laid out as a row ($F^{-1}$ is symmetric); in the toy, for $j = o$, the two numbers are $3/18 = 1/6$ and $-6/18 = -1/3$. Keeping the pair's block of $\delta(F^{-1})$,

$$\delta\mathsf{C}_w \;=\; -\big(F^{-1}\mathsf{A}\,F^{-1}\big)_{ww} \;=\; -\sum_j a_j\,(F^{-1})_{w,j}\,(F^{-1})_{j,w},$$

$$(\delta\mathsf{C}_w)_{\alpha\beta} \;=\; -\sum_j a_j\,(F^{-1})_{\alpha j}\,(F^{-1})_{j\beta},$$

where the sum writes the diagonal $\mathsf{A}$ out, column by column: each noisy column lowers $\mathsf{C}_w$ by $a_j$ times the outer product of its two-vector with itself (a column times a row, a $2 \times 2$ matrix).

How the determinant responds. $\ln \mathrm{FoM} = -\frac{1}{2} \ln\det \mathsf{C}_w$, and for a small change of a matrix $S$, $\delta \ln\det S = \mathrm{tr}\big(S^{-1}\,\delta S\big)$: the matrix version of $\delta \ln x = \delta x / x$, read off from $\det(I + X) \approx 1 + \mathrm{tr}\, X$ for small $X$. So

$$\delta \ln \mathrm{FoM} \;=\; -\tfrac{1}{2}\,\mathrm{tr}\big(\mathsf{C}_w^{-1}\,\delta\mathsf{C}_w\big),$$

with $\delta\mathsf{C}_w$ the block just computed.

From the trace to one number per column. Write the trace out in indices, in two moves. First the matrix product: its $(\gamma, \beta)$ entry sums over the shared middle index $\alpha$ ($\gamma$ is one more Greek index over the pair, since $j$ is taken),

$$\big(\mathsf{C}_w^{-1}\,\delta\mathsf{C}_w\big)_{\gamma\beta} \;=\; \sum_{\alpha} (\mathsf{C}_w^{-1})_{\gamma\alpha}\,(\delta\mathsf{C}_w)_{\alpha\beta}.$$

Then the trace: set the two outer indices equal, $\gamma = \beta$, and sum over $\beta$ – the diagonal entries of the product, added up:

$$\mathrm{tr}\big(\mathsf{C}_w^{-1}\,\delta\mathsf{C}_w\big) \;=\; \sum_{\beta} \big(\mathsf{C}_w^{-1}\,\delta\mathsf{C}_w\big)_{\beta\beta} \;=\; \sum_{\alpha\beta} (\mathsf{C}_w^{-1})_{\beta\alpha}\,(\delta\mathsf{C}_w)_{\alpha\beta}.$$

Finally insert the block above with its labels unchanged, $(\delta\mathsf{C}_w)_{\alpha\beta} = -\sum_j a_j\,(F^{-1})_{\alpha j}\,(F^{-1})_{j\beta}$:

$$\mathrm{tr}\big(\mathsf{C}_w^{-1}\,\delta\mathsf{C}_w\big) \;=\; -\sum_{j} a_j \sum_{\alpha\beta} (\mathsf{C}_w^{-1})_{\beta\alpha}\,(F^{-1})_{\alpha j}\,(F^{-1})_{j\beta}.$$

Each summand is a product of three plain numbers, so it can be reordered at will, $(F^{-1})_{j\beta}\,(\mathsf{C}_w^{-1})_{\beta\alpha}\,(F^{-1})_{\alpha j}$ – a chain $j \to \beta \to \alpha \to j$ in which each index meets its partner in the next factor – and the double sum over $\beta$ and $\alpha$ is then a row times a matrix times a column, $\sum_{\beta\alpha} x_\beta M_{\beta\alpha}\, y_\alpha = x^{\mathsf{T}} M y$, with $x^{\mathsf{T}}$ the row $(F^{-1})_{j,w}$ and $y$ the column $(F^{-1})_{w,j}$: exactly a quadratic form,

$$\mathrm{tr}\big(\mathsf{C}_w^{-1}\,\delta\mathsf{C}_w\big) \;=\; -\sum_{j} a_j\,(F^{-1})_{j,w}\,\mathsf{C}_w^{-1}\,(F^{-1})_{w,j},$$

$$\text{and with the } -\tfrac{1}{2} \text{ in front,} \quad \delta \ln \mathrm{FoM} \;=\; +\tfrac{1}{2} \sum_{j} a_j\,(F^{-1})_{j,w}\,\mathsf{C}_w^{-1}\,(F^{-1})_{w,j}.$$

In one sentence: $\delta\mathsf{C}_w$ is a sum of outer products, and the trace of a matrix times an outer product is the matching quadratic form, $\mathrm{tr}(M\,y\,x^{\mathsf{T}}) = x^{\mathsf{T}} M y$. The minus sign of $\delta\mathsf{C}_w$ cancels against the $-\frac{1}{2}$, and the statement is the familiar one: the noise shrinks the forecast ellipse and raises the FoM.

The trace step, checked in the toy. Let only the $o$ column be noisy, with the $a_o = 0.6$ of "Noise on the other column" above. The two-vector is $(F^{-1})_{o,w} = (1/6, -1/3)$, so the outer product gives

$$\delta\mathsf{C}_w \;=\; -0.6 \begin{pmatrix} 1/6 \\ -1/3 \end{pmatrix} \begin{pmatrix} 1/6 & -1/3 \end{pmatrix} \;=\; \begin{pmatrix} -0.0167 & 0.0333 \\ 0.0333 & -0.0667 \end{pmatrix}.$$

The trace side, with $\mathsf{C}_w^{-1} = F_w$ from part one: $\mathrm{tr}(F_w\,\delta\mathsf{C}_w) = -0.100$, so $-\frac{1}{2}\,\mathrm{tr}(\mathsf{C}_w^{-1}\,\delta\mathsf{C}_w) = 0.050$. The quadratic-form side: $(F^{-1})_{o,w}\,\mathsf{C}_w^{-1}\,(F^{-1})_{w,o} = 1/6$ (computed term by term in "$R_o^2$

in the toy" below, where Eq. (41) splits it as $R_o^2\,\sigma_o^2 = (1/3)(0.5)$), so $\frac{1}{2} \times 0.6 \times 1/6 = 0.050$: the two sides agree, as the reordering says they must.

Collected, with each column's number split into named factors,

$$\delta \ln \mathrm{FoM} \;\approx\; \frac{1}{2} \sum_j R_j^2\, a_j \sigma_j^2, \qquad R_j^2 \;=\; \frac{(F^{-1})_{j,w}\, \mathsf{C}_w^{-1}\, (F^{-1})_{w,j}}{\sigma_j^2}, \tag{41}$$

one term per column, each the two-column term $\frac{1}{2}\, a_j \sigma_j^2$ weighted by a number $R_j^2$ between 0 and 1.

What $R_j^2$ means. It is the squared multiple correlation of $p_j$ with the pair: the share of $p_j$'s forecast variance that a linear fit to $(w_0, w_a)$ explains. It is 1 for $w_0$ and $w_a$ themselves, recovering the first-order rule of Eq. (40), and 0 for a parameter uncorrelated with the pair: a nuisance column's noise reaches the FoM only through its correlation with the pair. For the tool, the $R_j^2$ are the source of the low cost: ranked by their weights $R_j^2 \sigma_j^2$, only a few heavy columns contribute, and only their noise needs measuring at each candidate setting.

$R_o^2$ in the toy. The two-vector is $(F^{-1})_{o,w} = (1/6, -1/3)$ and $\mathsf{C}_w^{-1} = F_w$, so the quadratic form is

$$\begin{pmatrix} 1/6 & -1/3 \end{pmatrix} \begin{pmatrix} 14/3 & 10/3 \\ 10/3 & 11/3 \end{pmatrix} \begin{pmatrix} 1/6 \\ -1/3 \end{pmatrix} \;=\; \begin{pmatrix} 1/6 & -1/3 \end{pmatrix} \begin{pmatrix} -1/3 \\ -2/3 \end{pmatrix} \;=\; -\frac{1}{18} + \frac{2}{9} \;=\; \frac{1}{6},$$

and dividing by $\sigma_o^2 = 1/2$ gives $R_o^2 = 1/3$. No pairwise correlation gives this number – squared, they are $0.30^2 = 0.09$ with $w_0$ and $0.53^2 = 0.29$ with $w_a$, and their sum 0.38 overshoots because $w_0$ and $w_a$ are themselves correlated – which is why the formula needs the full quadratic form. (A check without the formula: with the pair held fixed, $o$'s variance is $1/f_{33} = 1/3$, two thirds of its free variance $\sigma_o^2 = 1/2$; the pair explains the remaining third, $R_o^2 = 1/3$.) The first-order prediction for the toy's last case is then $\delta \ln \mathrm{FoM} = \frac{1}{2} \times \frac{1}{3} \times \frac{1}{2} \times 0.6 = 0.05$, a ratio of about 1.05 against the exact 1.041: the right size, a second-order distance away.

On the real system. Table 12 lists $R_j^2$ for every stencil column of the converged $\Delta z = 0.05$ system, for both published pairs at both cuts – and, in parentheses, the same numbers computed on a system built from the cheap $g = 64$ columns, for the recipe at the end of this subsection. Every cosmological column is strongly correlated with both pairs ($R_j^2 \approx 0.6$ to 1), so their noise reaches the FoMs almost undiminished, while the seven $\sigma_v$ columns stay low. The same formula also rates the per-bin columns: the $b_{1,i}$, degenerate with the amplitude $A_s$, reach $R_j^2 \approx 0.7$ with $(\Omega_m, 10^9 A_s)$ at $k_{\max} = 0.075\, h/\mathrm{Mpc}$.

**The part that averages to zero.** What the average misses. Eq. (41) keeps only the mean of the noise's effect. The middle term of Eq. (36), $2\, d_j^{\mathsf{T}} \mathsf{C}^{-1} e_j$, and the off-diagonal gains $e_j^{\mathsf{T}} \mathsf{C}^{-1} e_k$ have zero mean, but a forecast is run once, on one realization of the computed vectors, and in that one realization they are not zero. So the realized $\delta \ln \mathrm{FoM}$ is the average shift plus a zero-mean remainder, one term per column, linear in that column's noise.

How large the scatter is. Write $X = \sum_j R_j^2\, a_j \sigma_j^2$, twice the average shift of Eq. (41). Two limits bracket the scatter. A column's whitened noise spread evenly over the $N_d$ kept band powers overlaps the few directions the derivatives span only by chance, and the linear terms end with a standard deviation of about $(X/N_d)^{1/2}$ (without priors; priors only lower it). Noise concentrated along the derivative directions is the worst case: one column's linear term can reach the square root of that column's whole contribution to $X$.

| | FoM$(w_0, w_a)$ | | FoM$(\Omega_m, 10^9 A_s)$ | |
|---|---|---|---|---|
| column | $k_{\max} = 0.075$ | 0.2 | 0.075 | 0.2 |
| $10^9 A_s$ | 0.90 (0.87) | 0.74 (0.73) | 1 (1) | 1 (1) |
| $n_s$ | 0.79 (0.74) | 0.56 (0.56) | 0.97 (0.97) | 0.94 (0.94) |
| $H_0$ | 0.76 (0.76) | 0.65 (0.65) | 0.93 (0.93) | 0.89 (0.89) |
| $\Omega_b$ | 0.85 (0.77) | 0.64 (0.62) | 0.98 (0.97) | 0.92 (0.92) |
| $\Omega_m$ | 0.93 (0.89) | 0.84 (0.83) | 1 (1) | 1 (1) |
| $w_0$ | 1 (1) | 1 (1) | 0.94 (0.90) | 0.88 (0.87) |
| $w_a$ | 1 (1) | 1 (1) | 0.92 (0.87) | 0.82 (0.81) |
| largest $\sigma_v$ | 0.05 (0.05) | 0.12 (0.11) | 0.04 (0.04) | 0.19 (0.18) |

Table 12: The squared multiple correlation $R_j^2$ of Eq. (41) for each stencil column with the two FoM pairs, on the converged $\Delta z = 0.05$ system, at $k_{\max} = 0.075$ and $0.2\,h$/Mpc ($k_{\max}$ quoted at $z = 0$). A pair's own members have $R_j^2 = 1$ by construction. The last row is the largest of the seven $\sigma_v$ columns; the other six sit below it. In parentheses: the same $R_j^2$ computed on the cheap $g = 64$ system; the largest change is 0.08. Computed on the MCMC-settings system instead, the largest change grows to 0.15 ($\Omega_b$ at $0.075\,h$/Mpc).

Why the scatter wins near convergence. The average shift is built from the $a_j$, each proportional to the noise squared; the linear terms grow as the noise itself. Halve the noise and the shift falls fourfold, the scatter only twofold. So close to convergence the zero-mean scatter, not the average push, dominates the realized error.

The practical consequence. A quantity that scatters with the noise realization is not shown converged by shrinking a predicted mean; it is shown converged by checking that the realization does not matter. That is why the practical criterion below is independence of the step – two steps, two different noise realizations, the same FoM – and not the average formula alone: the cheap estimate picks the setting, the step test confirms it.

**The prediction against the measured offsets.** What the formula is for. Eq. (41) is the convergence diagnostic promised at the top of this subsection: it estimates, cheaply, at which setting the FoM stops changing appreciably – stabilizes. It does not have to predict the FoM's value; the FoMs of the science figures come from the converged derivatives, never from the formula, so order-unit accuracy on the shift is all the diagnostic needs.

What the comparison is. Table 13 compares, for every row of Table 9, the measured FoM ratio of Figure 4 with the ratio predicted from the cheap ingredients alone. Measured: FoM(configuration)/FoM(reference), the grid rows computed with the five-point rule at step $h = 0.02$, the reference the same rule on the finest grid of the starting configuration, $g = 512$. Predicted: $F$, $\sigma_j$ and $R_j$ come from the cheap $g = 64$ system (its stencil columns plus the starting configuration's per-bin columns), computed once; each row's noise gives its $a_j$ through Eq. (38), and the prediction is the exact FoM ratio of $F + \mathrm{diag}(a_j)$. The reference is not noise free – its columns carry the same noise floor as the $g = 256$ row, and on the cheap system the formula inflates it by 5.0% at $k_{\max} = 0.075\,h$/Mpc and 0.4% at $0.2\,h$/Mpc – so the quoted prediction divides each row's inflation by the reference's own. All fourteen stencil columns enter the prediction; the per-bin columns do not – their noise is not measured by Table 9, and swapping them moves the FoM by less than 0.1% ("The distance grid dominates" below), below the comparison's resolution.

| | $k_{\max} = 0.075\,h/\mathrm{Mpc}$ | | $k_{\max} = 0.2\,h/\mathrm{Mpc}$ | |
|---|---|---|---|---|
| configuration | measured | predicted | measured | predicted |
| FoM$(w_0, w_a)$ | | | | |
| MCMC settings | 1.556 | 5.98 | 1.130 | 1.935 |
| $g = 64$ | 1.292 | 2.13 | 1.059 | 1.170 |
| $g = 128$ (starting) | 1.006 | 1.202 | 1.004 | 1.024 |
| $g = 256$ | 1.000 | 1.000 | 1.000 | 1.000 |
| final settings | 1.021 | 0.969 | 0.995 | 0.997 |
| FoM$(\Omega_m, 10^9 A_s)$ | | | | |
| MCMC settings | 1.513 | 2.98 | 1.115 | 1.593 |
| $g = 64$ | 1.265 | 1.787 | 1.049 | 1.135 |
| $g = 128$ (starting) | 1.007 | 1.169 | 1.003 | 1.020 |
| $g = 256$ | 1.000 | 1.000 | 1.000 | 1.000 |
| final settings | 1.018 | 0.969 | 0.996 | 0.997 |

Table 13: Measured against predicted FoM ratios on $\Delta z = 0.05$. Measured: FoM(configuration)/FoM(reference) from Figure 4, the reference the $g = 512$ run of the starting configuration (five-point rule at step $h = 0.02$). Predicted, from cheap ingredients only – no converged run enters: the exact diagonal addition $F + \mathrm{diag}(a_j)$, with $F$, $\sigma_j$ and $R_j$ from the $g = 64$ system (its stencil columns plus the starting configuration's per-bin columns) and the $a_j$ from the matching row of Table 9, divided by the same prediction for the reference's own noise; the per-bin columns are omitted. $k_{\max}$ in $h/\mathrm{Mpc}$, quoted at $z = 0$.

How to read the table. Follow FoM$(w_0, w_a)$ at $0.075\,h/\mathrm{Mpc}$ down the grid rows: the predicted ratio falls $5.98 \to 2.13 \to 1.20 \to 1.00$ while the measured one falls $1.56 \to 1.29 \to 1.006 \to 1.000$. Both settle at 1 – the measurement by $g = 128$, the prediction one doubling later, at $g = 256$. That is the diagnostic doing its job, with a conservative margin: away from convergence the predicted shift is larger than the measured one, so it never declares a setting stable too early. The same reading holds at $0.2\,h/\mathrm{Mpc}$ and for FoM$(\Omega_m, 10^9 A_s)$. On the final-settings row both ratios lie within a few percent of 1, the stabilized level; the percent-level differences there come from the other controls that change with the final settings, below what the diagnostic is meant to resolve.

Why the margin is conservative. The additions pass whole to $F_w$ only for noise with no preferred direction in data space, while the realized error of the computed vectors is mostly a smooth, broadband shape within each spectrum – the kind of change the nuisance columns can fit. On the $0.075\,h/\mathrm{Mpc}$ cut at the MCMC settings, about 95% of the realized $w_a$-column error lies inside the span of the other columns, and the subtraction of Eq. (39) removes it; the formula counts all of the measured scatter, so its shift is the larger one – the safe direction for a convergence check. (At the MCMC settings the additions are also far beyond first order, $a_{w_a}\sigma_{w_a}^2 \approx 30$, Table 10, which is why the predictions use the exact diagonal addition throughout.)

A by-product: an upper bound. Read absolutely, the final-settings prediction bounds the residual noise inflation of the paper's FoMs from above: about 2.6% at $k_{\max} = 0.075\,h/\mathrm{Mpc}$ and 0.2% at $0.2\,h/\mathrm{Mpc}$, for FoM$(w_0, w_a)$ and FoM$(\Omega_m, 10^9 A_s)$ alike. The offsets themselves are measured, not predicted: the grid scan and the step test below do that.

A cheap check of a setting. The recipe behind the table's predicted column, usable long

before any converged run exists. Three steps. First, build one cheap system – here the $g = 64$ columns – and compute $F$, $R_j$ and $\sigma_j$ from it, once. Second, for each candidate setting measure the noise of the few heavy columns only: ranked by their contribution $R_j^2\, a_j \sigma_j^2$, two columns ($w_a$ and $w_0$) carry 90% of the prediction at $g = 128$, and four or five ($w_a$, $\Omega_m$, $H_0$, $w_0$, once one $\sigma_v$) at the final settings – four stencil evaluations per column, about 8 to 20 evaluations per candidate setting against 154 for a full derivative base. Third, evaluate the prediction and ask where it stops changing by order unity: there the FoM has stabilized – in Table 13, at $g = 256$, one conservative doubling after the measured $g = 128$. The cheap inputs are good enough for that question: recomputed with converged inputs, the predicted ratios move by at most 2% on the near-converged rows and 20% on the coarsest ones, and $R_j^2$ by at most 0.08 (the parentheses of Table 12) – immaterial for an order-unit diagnostic. And because the predicted shift is the conservative one, a setting the recipe accepts is at least as stable as predicted.

**What the estimator does and does not see.** The estimate treats the five errors as independent. An error that varies smoothly across the five points, a smooth offset of the whole stencil, cancels in the fourth difference the same way the smooth model does – and it cancels in the finite-difference derivative too, so it does no harm there either. The estimator thus measures exactly the part of the numerical error that hurts a derivative: the part that varies on the scale of the step. Its limit is the in-between case: if the errors at neighbouring points were partly correlated, the differences would partly cancel them, and the estimate would understate the jitter.

At the MCMC settings the $\Omega_m$, $w_0$ and $w_a$ columns are six to twelve times noisier than the $H_0$ column, which does not change with the grid. The finer grid quiets $w_0$ and $w_a$ most: the $w_a$ noise falls fortyfold from the MCMC settings to $g = 256$, the final grid.

**How the non-Limber spectra are computed: FFTLog on a distance grid.** The exact projection integrates each bin's radial kernel against spherical Bessel functions (Section 6.3; the formulas are in the summary note, `notes/bias_model_summary`, on the non-Limber spectra). The code (`cosmo2D.c`, `C_cl_tomo` and the low-rank core) computes it in three steps:

1. it samples each bin's radial kernels – the density, RSD and magnification rows, and in the low-rank path also the rank kernels – on a log-spaced grid in comoving distance $\chi$ covering $z = 0.002$ to 4;
2. FFTLog computes the Bessel transforms $F_\ell(k) = \int d\chi\, W(\chi)\, j_\ell(k\chi)$, and their $j_\ell''$ counterparts for the RSD rows: one forward FFT per row, done once, then the Hankel transforms in blocks of 16 multipoles;
3. $C_\ell = (2/\pi) \int dk\, k^2 P(k)\, F_\ell^i F_\ell^j$ is summed on the $k$ grid that FFTLog ties to the $\chi$ grid.

Everything therefore rests on how well the $\chi$ grid resolves the kernels.

**Where the grid factor enters.** The grid is endpoint-inclusive in $\log\chi$, with spacing $d\ln\chi = \ln(\chi_{\max}/\chi_{\min})/(N_\chi - 1)$ and $N_\chi - 1 = 511\, m$: the combined factor $g$, the product `accuracyboost` × `nonlimber_accuracyboost`, is rounded up to the power of two $m$ (at most 1024; `init_nonlimber_accuracy_boost` in `generic_interface.cpp`). Doubling $g$ halves every interval and keeps the old nodes, so a higher $g$ strictly refines the same grid, and "about $511\, g$ comoving distances" is exact whenever $g$ is a power of two.

**Why thin bins need a fine grid: a worked estimate.** The spacing is constant in $\ln\chi$, so a bin of fixed comoving width is narrowest in $\ln\chi$ at high redshift. Take the $\Delta d_{\rm com} = 62$ Mpc layout's ELG bin at $\bar{z} = 1.32$, with a Planck-like cosmology for the estimate: $\chi \approx 4{,}100$ Mpc, so the bin spans $\Delta\ln\chi \approx 62.3/4100 \approx 1.5\times10^{-2}$. Its Fingers-of-God-smoothed edges are far narrower: with $\sigma_v = 289$ km/s (the ELG2 fiducial), the edge width is $\sigma_\chi = (1+z)\,\sigma_v/H(z) \approx 4.6$ Mpc, or $\Delta\ln\chi \approx 1.1\times10^{-3}$. The grid spans $\ln[\chi(4)/\chi(0.002)] \approx 6.7$, so $d\ln\chi = 6.7/(511\,g)$:

- $g = 32$ (the MCMC value): $d\ln\chi \approx 4.1\times10^{-4}$, about 37 nodes across the bin and fewer than 3 across an edge;
- $g = 128$: about 150 across the bin, 11 across an edge;
- $g = 256$ (final): about 290 and 22.

At the MCMC grid an edge is sampled by two or three points. The likelihood records the consequence for the $\chi^2$ itself (`likelihood/desi_cl.yaml`): the smoothed edges of the highest-redshift thin bins resolve between combined factors 32 and 64, and below that the $z = 1.32$ ELG bin carries a coherent $0.18\sigma$ broadband amplitude bias.

**Why a coarse grid shows up as noise.** The grid's endpoints sit at fixed redshifts, $z = 0.002$ and 4, at the sampled cosmology. When a parameter changes the shape of $\chi(z)$, every kernel feature – a bin's edges above all – lands at a different position between the nodes, and with only a few nodes per edge the sampled kernel changes irregularly from one parameter value to the next. That irregular part is the $n(\theta)$ of "The noise of one evaluation". $H_0$ is the exception that proves the rule: it rescales every distance by one common factor, and on a grid log-spaced between fixed redshifts a common rescaling moves the nodes and the kernels together, so nothing lands differently. That is why the $H_0$ noise in Table 9 does not change with $g$, while $\Omega_m$, $w_0$ and $w_a$, which change the shape of $\chi(z)$, quiet down as the grid refines.

**The distance grid dominates.** Table 14 and Figure 4 give the offset of the FoMs from the converged value ($g = 512$), every control but the grid at the starting configuration's value. In the figure, follow the diamonds, the grid scan at step $h$, from left to right: at $0.075\,h$/Mpc the FoMs fall by 30 to 40% until $g = 128$ and then stay put. The coarse grids add noise that the FoM reads as information. The MCMC settings as a whole (the plus of the figure) raise FoM$(w_0, w_a)$ by 56% at $0.075\,h$/Mpc and by 13% at $0.2\,h$/Mpc, although their fiducial vector differs from the starting configuration's by only $\Delta\chi^2 = 0.015$. The grid alone accounts for most of it.

Which columns carry the offset? Swapping single columns (on the study's system) between configurations shows it: the $w_a$ column, then $w_0$ and $\Omega_m$. The per-bin and $\sigma_v$ columns move the FoM by less than 0.1%.

**The criterion: independence of the step.** The truncation error of a stencil grows with the step, as $\epsilon^4$ for five points, while the noise it divides falls as $1/\epsilon$. A derivative is converged when neither matters: when the FoMs, the errors and the shifts stay put as the step changes, every step scaled together, and as the stencil changes from three points to five.

The step markers of Figure 4, beside each grid, show the test. At $g = 128$ the five-point stencil at $h/2$ raises the FoM by 17%: the noise divided by a smaller step. At $g = 256$, from $h/2$

| grid factor $g$ | $k_{\max} = 0.075\,h/\mathrm{Mpc}$ | $k_{\max} = 0.2\,h/\mathrm{Mpc}$ |
|---|---|---|
| 32 (MCMC) | +40% | +12% |
| 64 | +29% | +5.9% |
| 128 (starting configuration) | +0.6% | +0.4% |
| 256 (final) | 0.0% | 0.0% |

Table 14: The offset of FoM$(w_0, w_a)$ of the paper model from the converged value ($g = 512$), on $\Delta z = 0.05$, every other control at the starting configuration's value. $k_{\max}$ quoted at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

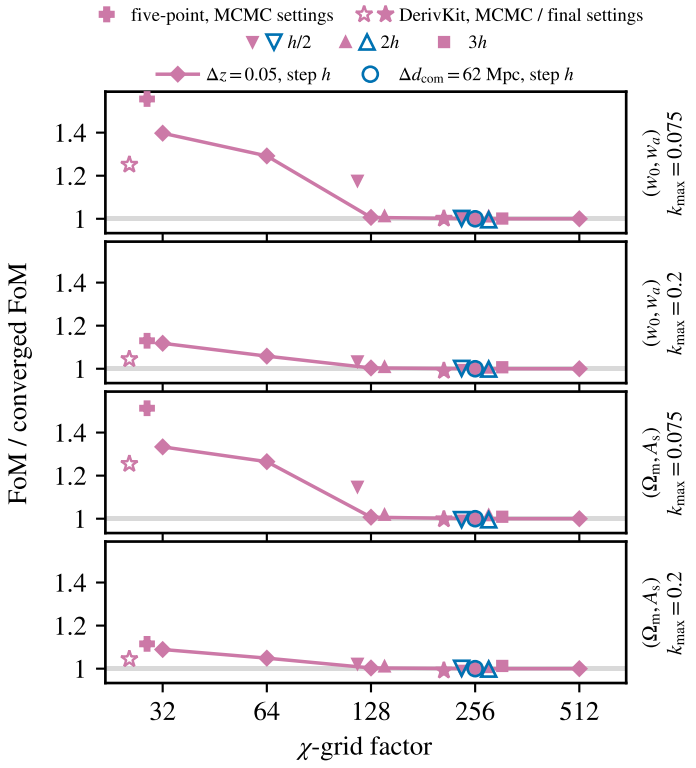



Figure 4: The paper model's FoMs on $\Delta z = 0.05$, as ratios to the converged value (the five-point stencil at step $h$ on the grid $g = 512$). From the top: FoM$(w_0, w_a)$ at $k_{\max} = 0.075$ and $0.2\,h/\mathrm{Mpc}$, then FoM$(\Omega_m, 10^9 A_s)$ at the same cuts. Diamonds: the grid scan at step $h$, every other control at the starting configuration's value. Down-triangle, up-triangle and square beside each grid: the steps $h/2$, $2h$ and $3h$. Plus: the MCMC settings. Stars: DerivKit, open at the MCMC settings and filled at the final settings. Open blue markers: $\Delta d_{\mathrm{com}} = 62$ Mpc at the final settings, steps $h/2$ and $2h$ against its own step-$h$ value (ring). Gray band: $1 \pm 1\%$. $k_{\max}$ quoted at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

to $3h$, both FoMs stay within 1.7%, the shifts move by at most $0.027\sigma$ (the study's system), and the three-point stencil agrees with the five-point one. The tolerance on the FoMs is about 1 to 2%; $g = 128$ agrees with $g = 256$ to 0.6% only at $h = 0.02$, so it stays a documented cross-check, and the final tier uses $g = 256$.

The same check on $\Delta d_{\rm com} = 62$ Mpc, at the final settings (the open blue markers of Figure 4, against the ring, that layout's own step-$h$ value): the five-point stencil at $h/2$ and $2h$ stays within 0.6% of step $h$ on both FoMs at both cuts, so $g = 256$ is step independent there too.

**CAMB's accuracy.** CAMB's `AccuracyBoost` scales its internal precision. The offsets of FoM$(w_0, w_a)$ from the converged value, with FoM$(\Omega_m, 10^9 A_s)$ in parentheses, on the study's system: at 1.05, the MCMC value, the FoM is 14.9% (16.6%) high; at 1.5, 2.0% (2.7%). Raising the boost from the starting configuration's 2 to 3 moves the FoM by 0.2% (0.5%). Every other control moves the FoM by at most 1.9%.

**The final settings.** The final tier (`DERIVATIVE_OVERRIDES` and `DERIVATIVE_CAMB` of the Fisher module) uses, on every layout:

1. CosmoLike's `accuracyboost` 3, which refines its tables in redshift, wavenumber and multipole, and `integration_accuracy` 1, the number of points of its Limber integrals;
2. the grid factor $g = 256$: `nonlimber_accuracyboost` 64, since $3 \times 64 = 192$ and the code rounds the factor up to the next power of two;
3. the low-rank rank 5 and multipole cap 1000, the values of the yamls;
4. CAMB's `AccuracyBoost` 2.0, `k_per_logint` 30 (wavenumbers per logarithmic interval of its output) and maximum wavenumber $60\,{\rm Mpc}^{-1}$, the likelihood's `kmax_boltzmann` 20 times `accuracyboost` 3;
5. the halofit patch below;
6. five-point stencils for every stencil and per-bin column.

Against the converged value ($g = 512$) on $\Delta z = 0.05$, these settings give FoM$(w_0, w_a)$ +2.1% at $0.075\,h$/Mpc and −0.5% at $0.2\,h$/Mpc, and FoM$(\Omega_m, 10^9 A_s)$ +1.8% and −0.4%. The errors $\sigma$ move by at most 1.6%, and the largest shift by at most $0.007\sigma$ (the study's system).

The remaining 2% sits in the $w_0$, $w_a$ and $\Omega_m$ columns together. At $0.075\,h$/Mpc these three parameters are nearly degenerate, and swapping any one of their columns alone moves the FoM by up to 3.5%, in opposite directions that largely cancel.

**DerivKit, the cross-check.** DerivKit is an external package, installed by Cocoa under `external_modules/code/derivkit`, that differentiates through a fit rather than a fixed difference. Along one parameter it

1. evaluates the vector at eleven points: the center $\theta_0$ and $\theta_0 \pm 1$, 1.5, 2.25, 3.375 and 5% of $\theta_0$. Below $|\theta_0| = 10^{-3}$ the steps become absolute, $10^{-6}$ to $4 \times 10^{-5}$, far below the noise; that is why the study samples a parameter whose fiducial value is zero ($w_a$, the $b_2$

offsets) through the shifted variable $u = 1 + \theta - \theta_0$, whose steps are 0.01 to 0.05 in $\theta$, the range of the five-point steps;

2. fits a straight line through them, entry by entry (`np.polyfit` of degree one for a first derivative), each point weighted by the inverse of its distance to the center. The fit squares the weights, so the center sample carries about a million times the weight of an edge point: the line is pinned to the center;

3. only when some relative misfit exceeds 5%, removes the worst-fitting point and its mirror – never the center – and refits, down to a floor of seven points. The data vector's noise is far below 5% of its values, so the first eleven-point fit is accepted, the pruning never runs, and the fallback `poly_at_floor` is never reached;

4. returns the slope of the line (the calls and defaults of `get_ddv_dkit` in `cosmolike_notebook_utils`, `min_samples` 7).

**What the fit really computes.** A line pinned to the center sample has a least-squares slope that reduces exactly to the plain average of the ten difference quotients $[m(\theta_i) - m(\theta_0)]/(\theta_i - \theta_0)$, and pairing each point with its mirror turns that into the mean of five central differences, one per step size. This one identity explains the method (Figure 5):

- **Noise.** The center's noise cancels between mirror points, and the average of five independent central differences beats any single one. In the toy of Figure 5, the spread of DerivKit's slope is 1.9% of the true slope against 4.8% for the five-point stencil, a ratio 0.39; the mean-of-five picture predicts the measured spread (1.88% both ways).

- **Bias.** A central difference at step $t$ errs by $m''' t^2/6$ (the even orders cancel between mirror points), so the mean keeps $(m'''/6)\,\langle t^2 \rangle$: a curvature bias the five-point stencil cancels by construction. In the toy it is +0.5%, the offset of the noise-free marker.

- **What no fit removes.** A smooth deterministic error, the signature of a coarse grid, moves all eleven points together: a bias, not a scatter. The toy's smooth ripple deflects all three estimators by the same 3.1 to 3.2%. A fit averages scatter; it cannot remove a bias.

The study applied DerivKit to the cosmological and $\sigma_v$ columns and to the $\{b_{1,i}\}$ and $\{b_{2,i}\}$ families, batch by batch (the stars of Figure 4).

- At the MCMC settings DerivKit's FoM is 25% high, against 56% for the five-point stencil: it reduces the noise but cannot remove the error of a coarse grid.

- At the final settings it agrees with the five-point stencil within 2.0% ($\sigma$ within 2.2% and shifts within $0.006\sigma$ on the study's system) and sits within 1% of the converged value ($g = 512$).

- It costs 11 evaluations per column against 4, about 210 s against 78 s per column.

The final tier therefore uses five-point stencils everywhere, with DerivKit as the documented cross-check.

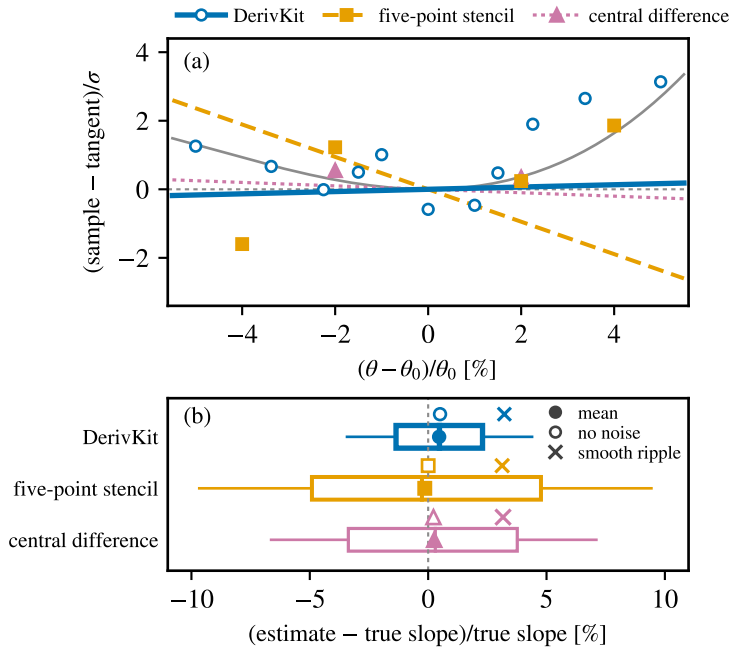



Figure 5: A toy comparison of the three derivative estimators, not a measurement: $m(\theta) = 1 + t + 0.8\,t^2 + 5.6\,t^3$ with $t = \theta - \theta_0$ and $\theta_0 = 1$, so the true slope is 1; independent Gaussian noise of $\sigma = 10^{-3}$ at every evaluation; and the estimators called as the study calls them: a central difference at $\pm 2\%$ of $\theta_0$, the five-point stencil at $\pm 2$ and $\pm 4\%$, and DerivKit at the center and $\pm 1$, 1.5, 2.25, 3.375 and 5%. (a) One typical realization: each estimator's samples minus the true tangent at $\theta_0$, in units of $\sigma$, with its fitted slope drawn through the center; the gray curve is the true curve minus the tangent. (b) The error of the estimated slope over 2,000 realizations: the box spans the 16th to 84th percentiles, the whiskers the 2.5th to 97.5th, the filled marker is the mean, the open marker the noise-free estimate (the curvature bias), and the cross the error on a smooth deterministic ripple, $10^{-3}\sin(2\pi t/0.2)$, added without noise, which every estimator follows alike.

**The CAMB halofit patch.** Halofit is CAMB's fitting formula for the nonlinear matter power spectrum. It finds the nonlinear scale by bisection (repeated halving of an interval) on $\sigma(R)$, the rms linear density contrast smoothed on the scale $R$; stock CAMB stops at $|\sigma(R) - 1| \leq 10^{-3}$. Where the bisection stops jumps between nearby cosmologies and shifts the nonlinear power spectrum by about 0.3%, a noise that a finite difference divides by its step.

Cocoa's installation key `PATCH_CAMB_HALOFIT_TOLERANCE`, on by default, tightens the test to $10^{-7}$. The final tier requires it: the builders stop at start unless `halofit.f90` holds the $10^{-7}$ test and the compiled CAMB library is newer than that file (`check_halofit_patch`).

**Two tiers.** The project fixes two sets of accuracy settings, two tiers, which `derivative_settings` returns per layout.

- **Exploration**: fast settings for quick looks; its FoMs are not converged.
- **Final**: the settings above, for every figure and number of a paper or of a note's final

version. The criterion is the stability of the FoMs, the errors and the shifts; runtime is not a criterion.

Both tiers use five-point stencils for every stencil and per-bin column, and exact templates for $\{\mathrm{SN}_{0,i}\}$ and $\{\mathrm{SN}_{2,i}\}$. A cache made with central differences belongs to neither tier, and the builders refuse it.

Each cache records its tier, its settings, and a fingerprint (`settings_fingerprint`): a 64-character code (a sha256 hash) computed from the full model input, so that any change of the input changes the code. The input is the example yaml merged with the likelihood's options and parameter files, the accuracy settings, the priors and the ties between parameters, together with the fiducial point and the tier. Unlike a comparison of file names, the fingerprint changes when a yaml is edited under the same name, and stays the same when a file only moves.

**The two routes agree.** The study computes its derivatives by a second route: it unties the per-bin parameters and steps them through the full likelihood, with no step class (Appendix A). The agreement study (`tests/studies/results/fisher_derivatives_agreement_dz0p05.json`) compares that route with the production cache of $\Delta z = 0.05$ at the final tier, column by column. The per-bin $b_{2,i}$, $b_{s^2,i}$ and $\alpha_{0,i}$ columns agree exactly (relative difference zero); the $b_{1,i}$ columns, which the two routes connect through the chain rule of Eq. (13), agree to $5.2 \times 10^{-11}$; and the cosmological and $\sigma_v$ stencil columns to $4.6 \times 10^{-14}$.

## 10 Limits and checks

**What needs new evaluations of the data vector.**

- **A physical term outside the base.** A term that is no combination of the knobs, such as a nonzero third-order bias or redshift-space one-loop terms (which CosmoLike lacks), needs new base columns. A new combination of existing knobs, such as a quadratic tilt or another calibration slope at the same fiducial values, is only a new $\mathsf{M}$.

- **A different fiducial point.** Every derivative is taken at $p_0$. A new fiducial cosmology, new fiducial biases, or a calibration law that moves the fiducial $b_{2,i}$ needs new base columns, and a new covariance, which sits at the one-loop fiducial. A new covariance alone needs no evaluation: the notebook rebuilds every $\mathsf{G}_m$ from the current covariance file at every run.

- **Dependence beyond first order.** The forecast is the curvature at $p_0$, and Eq. (25) is first order in $\delta m$. The band powers are quadratic in the biases, so a large truth, such as an offset of twice a prior width or the 22% BGS flux-limit drift, is treated to first order, and a shift near the $\sigma/3$ threshold needs a refit, with new evaluations, to be confirmed.

**The checks that guard the machinery.**

- **The chain rule.** The per-bin columns, obtained through the step classes and summed over a group, $\sum_{i \in g} \partial m / \partial \beta_i$ with Eq. (13), must equal the stencil derivative $\partial m / \partial B_g$, obtained through the full likelihood. The two routes share no evaluation; the notebook's own check finds they agree to at most $1.5 \times 10^{-9}$ of the stencil column's largest entry on

every layout (the truncation error of the two stencils), and the final-tier agreement study of Section 9 tightens that to $5.2 \times 10^{-11}$ on $\Delta z = 0.05$.

- **Cross-cache checks.** The order-0 templates of the template cache equal the derivative cache's $\mathrm{SN}_{0,i}$ templates, and the two caches' fiducial vectors are equal. $\mathsf{B}$ carries the difference of the two vectors as its check column, so their $\Delta\chi^2$ is a diagonal entry of $\mathsf{G}$ on the shipped mask; it is zero on every layout. The vector after the last step equals the fiducial one bit for bit (the `reset` entry): no step survives its evaluation.

- **The matched-prior identity.** When the truth's linear-bias scatter equals the fit's prior and every other parameter is drawn from the fit's own priors, the scatter and noise terms of Eq. (26) must add up to exactly $n_{\mathrm{dof}}$: the widening the prior gives the ellipse equals the error the scatter causes (the correlated-prior note, on the test statistic).

- **One answer from independent constructions.** A + and a − truth give shifts of equal size (measured 0). Section 9 of the notebook builds the paper model's $\mathsf{M}$ by its own route, and at $\sigma_v$ width factor 1 it reproduces the $\alpha_2$-fixed fit of section 11 exactly; the pair-threshold scan with every pair kept reproduces the FoMs of sections 6 and 9 exactly. Each agreement tests $\mathsf{G}$, $\mathsf{M}$, the priors and the pseudo-inverse together.

- **The cache check.** Each cache stores its settings, tier and fingerprint, and the notebook refuses a cache whose entries differ from its own (Section 11).

## 11 Using the caches

The caches are the pickle files of Table 8, one set per layout, in `chains/`, which git ignores: a fresh clone recomputes them (hours), and every later run reads them. Their names start with `EXAMPLE_EVALUATE_FISHER`, and $\langle n \rangle = 1$ to 4 numbers the layouts in the order $\Delta d_{\mathrm{com}} =$ 62 Mpc, $\Delta z = 0.02$, 0.05, 0.2.

- `_MODEL_SCALE_CUTS.pkl`, written by the notebook's section 3: the bands each ultraviolet-fraction cut keeps, computed on the paper fiducial, at $k_{\max} = 0.05, 0.075, \ldots, 0.25\,h/\mathrm{Mpc}$, with its settings and fingerprints; the scans use the seven cuts up to 0.2. (The linear notebook, `EXAMPLE_EVALUATE_FISHER.ipynb`, keeps its own cut cache on the linear-bias fiducial for the linear-bias experiments.)

- `_MODEL_DERIVATIVES_<n>.pkl` and `_MODEL_TEMPLATES_<n>.pkl`: the base caches of Table 8 – the stencil and per-bin operator columns, the $\mathrm{SN}_{0,i}$ templates, the fiducial vector and biases in the first; the exact $\alpha_2$ columns and the $\mathrm{SN}_{0,i}$ and $\mathrm{SN}_{2,i}$ templates in the second – written by the notebook's section 4 or by `scripts/compute_fisher_base_mpi.py`, which writes the same files bit for bit.

Neither the notebook nor the script caches a Fisher matrix, $\mathsf{G}_m$ or FoM: these depend on the covariance and are rebuilt at every run.

**Explicit compute, save and load cells.** Caches are computed, saved and loaded only in their own cells. The compute cells run only with their switch set by hand (`COMPUTE_SCALE_CUTS` for section 3, `COMPUTE_BASE` for section 4, both `False` by default); the layouts are

computed cheapest first. The save cells refuse to replace an existing file unless called with `overwrite=True`, and write both caches of a layout or neither. The load cells raise when a cache is missing (naming the switch and the MPI script) or when its stored settings or fingerprint differ from the notebook's own: a mismatched cache is never read past the check. The MPI script follows the same rules with `--overwrite`, and stops without writing when its ranks see different files or a file changes during the run.

**What makes a cache out of date, so it must be recomputed.**

- **Detected automatically**: any change of the model input that the fingerprint covers (a likelihood option, a CAMB argument, a prior, a tie between parameters, the fiducial point), of the tier, of the stencil rule or of the steps. The load cell stops and names the mismatch. A file that moves or is renamed with the same content keeps its cache.
- **Not detected, so recompute by hand**: a change of the code itself (the likelihood's Python, CosmoLike, CAMB beyond the halofit check), since the fingerprint covers the input, not the code. No check compares a cache with the current code unless a cache is recomputed; the fiducial-vector check then catches a change that moves the fiducial vector.
- **Needs no recomputation**: a new covariance, a new prior or prior width, a new model built from the cached knobs, and any cut already in the cut cache. A $k_{\max}$ outside it needs the cut compute cell for that cut (CosmoLike's split of each band into the power below and above the cut), but no new derivative.

# A The per-bin step classes

**Why a step needs the likelihood's arrays.** The sampled parameters offer no per-bin handle: the parameter files tie each bin's linear bias to its group amplitude, and the likelihood computes $b_{2,i}$ and $b_{s^2,i}$ from $b_{1,i}$ through the calibrations, recomputing and resending every per-bin value at every evaluation. The completed per-bin values appear in one place: the argument lists of the compiled setters through which the likelihood hands them to CosmoLike (`set_nuisance_bias_tidal` for $b_{1,i}$, $b_{2,i}$, $b_{s^2,i}$ and the slot that carries $\alpha_{0,i}$; `set_nuisance_bias_alpha2` for $\alpha_{2,i}$). The batch steps of Section 6.2 are applied there. Resending the unchanged values does no harm; a step only has to change the entries of the batch's bins and leave everything else as it is.

**A subclass of the likelihood.** The notebook and the MPI script evaluate each layout on a subclass of its paper likelihood, defined in `scripts/compute_fisher_base_mpi_classes.py`: the mixin `PerBinStepMixin` listed before the likelihood class in the subclass's bases (for $\Delta d_{\rm com} = 62$ Mpc, `desi_cl_model_b_steps`). A mixin is a class never used alone: its methods run first and reach the likelihood's own through `super()`. It adds no option and no parameter – only three methods a caller uses:

```
like.set_per_bin_step(operator='b2', bins=members, step=0.02)
vector = model.dv()        # the point with the step added
like.clear_per_bin_step()  # the next point is the likelihood's own
```

`operator` is one of `b1`, `b2`, `bs2`, `a0` and `a2`, `bins` the 0-based bins of the batch, and `step` the signed step. The set and the clear are wrapped in `try/finally` (`evaluate_with_step` in the notebook), so no error leaves a step installed. Without a step the subclass evaluates exactly what its parent does.

**The stepped evaluation.** The likelihood's `set_lens_related` computes a point's per-bin arrays and sends them to CosmoLike. The mixin overrides it: the parent runs in full – every array, the Fingers-of-God $n(z)$, every setter – and then, with a step set, the stepped operator's setter is called once more with the step added (trimmed):

```
def set_lens_related(self, **params):
    super().set_lens_related(**params)   # the point's own arrays
    if self._step is None:
        return
    # fresh copies of the arrays the parent just sent
    B1, B2, BS2, BK = ...
    if operator == 'b1':
        B1[members] *= 1.0 + step
    elif operator == 'b2':
        B2[members] += step
    elif operator == 'bs2':
        BS2[members] += step
    elif operator == 'a0':
        # alpha_0 k^2 P = 2 b1 bK k^2 P with k in H0/c units
        BK[members] += step/(2.0*B1[members]*C_OVER_H0**2)
    ci.set_nuisance_bias_tidal(B1=B1, B2=B2, BS2=BS2, ..., BK=BK)
```

`B1`, `B2` and `BS2` hold $b_{1,i}$, $b_{2,i}$ and $b_{s^2,i}$, and `BK` the nonlocal bias $b_{K,i}$, one entry per bin. The $b_{1,i}$ step is relative, the $b_{2,i}$ and $b_{s^2,i}$ steps absolute, and an $\alpha_{0,i}$ step enters `BK`, because CosmoLike writes the counterterm of Eq. (3) as $2\,b_{1,i}b_{K,i}\,k^2P_{\rm mm}$ with $k$ in units of $H_0/c$ (`C_OVER_H0` $= c/H_0$ in Mpc/$h$). An `a2` step instead calls `set_nuisance_bias_alpha2` with the step added to the point's $\alpha_{2,i}$ on the batch's bins. Each compiled setter stores a value only when it differs from the stored one by more than a relative $10^{-13}$, then renews the cache key of the bias tables and computes nothing else, so the second call leaves CosmoLike in the state that one call with the shifted arrays leaves it in.

**The model stays the paper yaml's.** Cobaya finds the subclass through the `class` and `python_path` keys of a likelihood block. `step_yaml` writes them into a temporary copy of the paper yaml; a cobaya class without a yaml file of its own inherits its parent's defaults (options, parameter files, dataset), so every model detail still comes from the parent. `check_step_yaml` stops unless the copy resolves to the same cobaya input as the paper yaml – the likelihood options, theory, parameters, priors and fiducial point, compared blockwise.

**The batch loop.** The loop of the notebook's `fisher_base` walks the four stenciled families and the four batches (trimmed; `OPERATOR_STEPS` holds the steps $\epsilon$ of Section 6.2: 0.01 for $b_{1,i}$, 0.02 for $b_{2,i}$ and $b_{s^2,i}$, $5\,h^{-2}\,\mathrm{Mpc}^2$ for $\alpha_{0,i}$):

```
for op, step in OPERATOR_STEPS.items():
    Dop = np.zeros((len(m0), nbins))
    for c in range(max(color) + 1):
```

```
        members = np.array([i for i in range(nbins) if color[i] == c])
        vectors = []
        for offset in fm.FIVE_POINT_OFFSETS:
            vectors.append(
                evaluate_with_step(model, like, op, members, offset*step))
        d = fm.five_point(vectors, step)
        for i in members:
            # the b1 step is relative: (b1 dm/db1)/b1 = dm/db1
            Dop[rows[i], i] = d[rows[i]]/(b1[i] if op == 'b1' else 1.0)
    ops[op] = Dop
```

`color[i]` is bin $i$'s batch number from the greedy rule of `pair_coloring` (the code keeps the mathematicians' word for a batch); `fm.FIVE_POINT_OFFSETS` $= (2, 1, -1, -2)$ are the offsets $o$ of Eq. (14), and `fm.five_point` combines the four vectors with the stencil weights and divides by the step. `rows[i]`, built once by `spectrum_rows`, lists the entries of every spectrum that contains bin $i$; column $i$ receives the derivative on those rows and stays exactly zero elsewhere, so the zeros of Table 3 are never computed. The $b_{1,i}$ column is divided by $b_{1,i}$, because its relative step makes the stencil return $b_{1,i}\,\partial m/\partial b_{1,i}$. The $\alpha_2$ loop is the same with two evaluations per batch (Section 6.3). The MPI script evaluates the same points with the same classes and combines them with the same arithmetic; one rank runs them in this order.

**The caches.** Every stencil evaluation is at the same sampled point; the step rides inside the likelihood. Cobaya (the framework that runs the likelihood) stores the previous point's result and would return it without calling the likelihood, so `ModelFisher.dv` (`interface/cosmolike_lssty1xdesi_fisher.py`) calls `self.model.loglike(point, ..., cached=False)`, and the likelihood, with its step, runs at every evaluation.

CosmoLike keeps caches of its own, one key per sector of inputs (`cosmolike_core`'s `README.md`, section "The sector-ladder cache check"). A setter compares every incoming value with the stored one (`fdiff` in `generic_interface.cpp`) and, when any differs, draws a fresh random number as the sector's key (`nuisance.random_galaxy_bias` for the biases); every cached projection stores the keys it was built under and recomputes on a mismatch (`cosmo2D.c`). A batch evaluation therefore rebuilds the bias-dependent results and reuses everything keyed on other sectors – CAMB's power spectra, the distances, the power and FAST-PT tables. On $\Delta d_{\rm com} = 62$ Mpc at the final settings, an evaluation whose only change is the biases takes about 36 s, against about 2.5 minutes at a new cosmology (the laptop of Section 6.4). The return to the fiducial values is one more change, caught by the same comparison, and clearing the step needs no cleanup: the likelihood resends every per-bin value at the next point anyway.

**What tests this.** The sector-key machinery is tested by the sector-ladder cache check, `tests/data_vector/test_cache_consistency.py` (`tests/data_vector/README.md`, section "The sector-ladder cache check"): a deterministic ladder of mixed cosmology, bias and Fingers-of-God updates on all four layouts, a scramble of every sector at once, and bitwise assertions, with a second model walking the mirrored ladder. That test runs the frozen linear-bias configuration, so it exercises the linear setter `set_nuisance_bias`, not the one-loop `set_nuisance_bias_tidal` (Appendix B). The stepped route itself is validated against derivatives that never pass through a step class: the notebook's chain-rule check reproduces the group-amplitude

stencils from the per-bin columns through Eq. (13), to at most $1.5 \times 10^{-9}$ of the group column's largest entry over the four layouts, and the route-agreement study of Section 9 unties every per-bin parameter and steps it through the full likelihood: its $b_{2,i}$, $b_{s^2,i}$ and $\alpha_{0,i}$ columns agree exactly with the stepped ones, the $b_{1,i}$ columns through the chain rule to $5.2 \times 10^{-11}$. The reset check of Section 10 confirms, bit for bit, that no step survives its evaluation.

# B Open issues

- **The sector-ladder test has no one-loop rung.** The cache-consistency ladder exercises the linear setter `set_nuisance_bias` only; a ladder whose bias updates run through `set_nuisance_bias_tidal` (offset, $\alpha_{0,i}$ and $\mathrm{SN}_{0,i}$ steps) is still to be added.

- **Some accuracy offsets are scored on the study's system.** The error and shift stabilities of Section 9 (the $0.027\sigma$ step stability, the 1.6% of the errors and $0.007\sigma$ at the final settings, DerivKit's 2.2% and $0.006\sigma$), the CAMB-boost and other-control offsets and the column-swap attribution are scored on model B + tilts with $\alpha_2$ free and the exploration $\sigma_v$ widths, the convergence study's own system; only the FoMs were rescored for the paper model on the paper-fiducial cuts. The derivative columns are shared, so a rescoring would change the quoted stabilities, not the chosen settings.

# References


[1] T. Lazeyras, C. Wagner, T. Baldauf and F. Schmidt, "Precision measurement of the local bias of dark matter halos", JCAP 02 (2016) 018, arXiv:1511.01096.

[2] M. Maus et al., "An analysis of parameter compression and full-modeling techniques with Velocileptors for DESI 2024 and beyond", 2024, arXiv:2404.07312.

[3] E. Saraivanov, H. Rubira, V. Miranda and T. Eifler, "Towards the Two-Loop EFTofLSS in Galaxy Lensing Surveys", 2026, arXiv:2603.13031.

[4] K. B. Petersen and M. S. Pedersen, "The Matrix Cookbook", Technical University of Denmark, 2012.